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A rapid systematic review and case study on find, test, trace, isolation and support policies for COVID-19 prevention and control.

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Title: A rapid systematic review and case study on find, test, trace, isolation and support policies for COVID-19 prevention and control

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ABSTRACT

Objective To conduct a rapid review on the efficacy and policies of Find, Test, Trace, Isolate, Support (FTTIS) in COVID-19 prevention and control.

Setting and participant studies Research articles and reviews on the use of contact tracing, testing, self-isolation and quarantine for COVID-19 management published in English within 1 year (May, 2019 to 28th May, 2020) were eligible for the review. We searched MEDLINE (PubMed), Cochrane Library, SCOPUS and JSTOR with search terms including "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title or abstract. Studies not associated with FTTIS or COVID-19 or being solely commentary were excluded.

Outcomes A narrative synthesis with a tabulation system was used to analyse studies for their diverse research designs, methods, and implications.

Results Among the 163 initial publications, 50 eligible studies were included in the review. Included studies applied various designs: experiments, clinical studies, Government Documents, systematic reviews, observational studies, surveys, practice guidelines or technical reports. We identified in our systematic review the core elements for an effective Find, Test, Trace, Isolate, Support (FTTIS) system to interrupt the spread of a novel infectious disease, where treatment or vaccination are yet available, such as the current COVID-19 pandemic. We reported optimal strategies in the literature to shorten case finding time, improve both accuracy and efficiency of the test, coordinate stakeholders and sectors involved in a FTTIS system, support individuals to follow FTTIS and appropriately use digital tools to facilitate the program.

Conclusions Results of the review may inform countries considering implementing these measures.

Article summary

What is already known on this subject?

As treatment or vaccination are yet available, COVID-19 prevention and control depend on effective case finding, quarantine and isolation.

Strengths and limitations of this study

- The study systematically review the core elements for an effective Find, Test, Trace, Isolate, Support (FTTIS) system to interrupt the spread of COVID-19. Results of the review may inform countries considering implementing these measures.
- We reported optimal strategies in the literature to shorten case finding time, improve both accuracy and efficiency of the test, coordinate stakeholders and sectors involved in a FTTIS system, support individuals to follow FTTIS and appropriately use digital tools to facilitate the program.
- The limitation of the study is the qualitative review, as there were too few quantitative studies in the subject area to enable a meta-analysis.

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INTRODUCTION

In the absence of an effective vaccine, interruption of the transmission of COVID-19 depends on rapid isolation of infected individuals. For this to happen, a complex set of co-ordinated elements must be put in place to Find potential cases, either by identifying those who have symptoms or screening of individuals in high-risk settings, Test to confirm the presence of infection, Trace contacts, Isolate those infected and their contacts,[1] and Support[2] those in isolation to reduce the risk that they will breach any restrictions.

While various elements of a Find, Test, Trace, Isolate, Support (FTTIS) system have long been core elements of the public health role, COVID-19 has some specific characteristics, such as transmission from presymptomatic individuals[3] and a strong age gradient in both likelihood of infection and disease severity, as well as others that remain poorly understood. Despite this uncertainty, successful COVID-19 FTTIS programs have been reported to display coordination and smooth teamwork between the essential sectors, i.e. central and local government authorities, laboratories and the medical care system.[1] The sectors are interconnected by integrated and real-time information flow. However, methods to achieve this integral system remain inconclusive and largely determined by the existing organisations and infrastructures.

Digital tools are proposed to enhance the efficacy of FTTIS programmes.[1,3] Although the pilot study of the NHS contact tracing App showed positive results in decrease of the rate of transmission,[4] challenges remain in its coverage and general usage for it to be effective.[5] Individuals or occupation groups with high COVID-19 risk will need additional support to access the programme[2] or reduce burden of unnecessary or repeated episodes of quarantine. Other challenges for adopting a FTTIS programme include optimal strategy to shorten time to find the case and contacts,[1] accuracy of the test,[1] funding and staffing of the system,[6] barriers in adhering to FTTIS regulations,[7,8] and perceived economic impact on the livelihood of individuals.[8] To inform and support the design, implementation and betterment of a FTTIS system, we have undertaken a rapid review and structured it to answer a series of key questions for a successful FTTIS.

METHODS

The systematic review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. The review protocol is described as the following.

Eligible studies are research articles and reviews evaluating the effectiveness of contact tracing, testing, self-isolation and quarantine on COVID-19 management published in English within 1 year (May, 2019 to May, 2020).

Search We searched MEDLINE (PubMed), Cochrane Library, SCOPUS and JSTOR. The search terms included "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title or text. The date last searched is 28th May 2020. Studies were limited to those in which the subjects were human. Relevant reports and literature cited in the papers identified, as well as material from other sources such as the COVID Response Monitor developed by the European Observatory on Health Systems and Policies were also considered.[9]

Study selection In the MEDLINE search, clinical studies, clinical trials, evaluation studies, government documents, journal articles, meta-analyses, reviews, systematic reviews, multicentre studies, observational studies, practice guidelines, pragmatic clinical trials and technical reports were studied. Studies whose subject area were in medicine or social sciences in SCOPUS, and research reports in JSTOR (excluding book chapters) were selected. We screened title, abstract and content to remove studies unrelated to FTTIS or COVID-19.

Following discussion with individuals involved in the COVID-19 response in the UK, a list of key questions to answer was developed and answered using the literature collected (Supplementary section one). For each

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publication we extracted (where appropriate) information on study design, method, results, main findings, and limitations using the PICOS tool.[10] Risk of bias was assessed where appropriate.

Patient and Public Involvement

There is no patient and public involvement in the study.

RESULTS

We identified 50 studies from PUBMED (Medline), 43 studies from SCOPUS, 56 from JSTOR, 1 from the Cochrane library and 4 from additional sources. The flow diagram is set out in Figure 1. From the 164 initial publications, we removed 30 duplicate studies, 84 items that were not directly focused on COVID-19 or strategy of testing, contact tracing or isolation, or not in the format of a research report (such as editorials or podcasts). We thus included a total of 50 studies in the review.

Overview of FTTIS

To set the scene, we begin with a non-exhaustive review of some key findings on the COVID-19 pandemic. A review by Nussbaumer-Streit B et al. concluded that a combination of quarantine and other nonpharmaceutical interventions would significantly reduce COVID-19 incidence and mortality.[11] Policy decisions must be informed by a detailed understanding of transmission dynamics. With growing evidence that so-called super-spreading events play an important role in many outbreaks,[12] that finding and testing those present at such events is a priority. This relates to work by Manchein et al who have proposed that the growth of COVID-19 cases follows a power-law pattern rather than an exponential one, suggesting that transmission is characterised by features of scale free networks, fractal kinetics, and small world features. The implication is that while social distancing has been important to slow the spread of COVID-19 so far, what they describe as a "huge" expansion of testing and isolation will be necessary to permit economies to reopen. These conclusions are consistent with work from Italy stressing the importance of maintaining very strict quarantine of those affected. [13] Policies must also be underpinned by high quality data, yet some countries have so far struggled to put well-functioning integrated systems in place. Inadequate testing will often underestimate the true number and growth rate of COVID-19 cases.[14] This is a problem because testing is at the heart of a comprehensive FTTIS system. It is the only way to make a diagnosis where symptoms are atypical or unclear or in cases that are asymptomatic or presymptomatic.[15] There are now a number of reports on how FTTIS systems have been designed and operated.[16]

1. Infrastructure and procedure of a sampling-to-results process that is 24 hours or less.

Countries have differed in whether they have built on existing infrastructure or created new structures to deal with the pandemic. We have identified two modalities from the reviewed studies, one is a patient-based and the other a laboratory network. The UK has largely adopted the former approach and Hill et al. reported on the operation of a drive-through SARS-Cov2 testing in the NHS in Scotland[17] with patients being informed of their results within 24 to 36 hours. However, this service could only be used by those with access to a private vehicle and who were fit to drive. Lagier et al reported on SARS-Cov2 RT-PCR testing of repatriated French citizens.[18] Samples from 337 passengers on the repatriation flights were tested and the time from sample-to-result was 4h 50 min (6h 40min with the first returning flight). Their paper describes how the laboratory optimised the testing strategy by prioritising extraction of RNA from samples, reducing the time from sample to results to 2 h 50-55 minutes.

Binniker has set out the steps that should be taken when scaling up activity in existing laboratories,[19] which has the advantage of using existing networks, including transport and communications. Similarly, Gupta et al have described the response of laboratories in universities and clinics in India being converted to testing

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centres.[20] Tolia et al described the implementation of a testing programme within a hospital emergency department.[21]

Strategies to improve the testing response

The case definition of COVID-19 changed frequently in the initial stage of the epidemic, as knowledge of the disease evolved. In the early stage of the epidemic, test kits and laboratory capacity were in short supply in many countries.[22] To investigate the potential SARS-Cov-2 local transmission, the Dutch National Outbreak Management Team conducted a rapid screening among health care workers in an area of suspected COVID-19 outbreak. Testing followed a national protocol and was carried out either locally or in central laboratories. Nine hospitals were asked to sample health care workers on Saturday, 7th March and results were due on Monday, 9th March. When the SARS-Cov-2 positive rate came back as 4.1% among 1,097 tested, the local outbreak was confirmed.

Other laboratory-based strategies to improve testing efficiency include Yan et al. suggesting placing samples in reagents containing guanidine salts, for example TRIZOL, TRIZOL LS, or AVL buffer, to inactivate the virus and protect RNA.[23] They also, in situations where laboratory capacity is limited, suggested that samples could be pooled, with only those samples contributing to pooled batches then tested individually.[24] The authors proposed a means to calculate optimal sizes of pools derived from the prevalence rate. A similar three-stage pooling system was reported by Eberhardt et al who showed that it was possible to test three and seven times as many individuals with the same number of test kits when prevalence rates were 5% and 1%, respectively.[25]

Minimising false testing results.

False testing results can impede both the treatment of the patient and public health interventions to contain infection. The testing accuracy may be influenced by sampling and testing kit performance. The gold standard for COVID-19 diagnosis remains rRT-PCR–based assays with respiratory specimens.[26] Among different respiratory sampling methods, rRT-PCR assays based on nasopharyngeal swabs tend to have lower cycle threshold than oropharyngeal swabs, indicating a higher viral load collected through nasopharyngeal swabs.[23] Zitek reported individuals who are initially RT-PCR negative can become positive with repeated tests.[27] The authors attribute this to differences in the amount of virus at the site samples and viral shedding at the time of sampling.[27,28] Sampling error can occur due to low levels of virus at beginning or end period of shedding and inadaquate swabs. Thus, they advocate taking multiple samples,[27,28] and supporting rRT-PCR results with Chest Computerized Tomography imaging for COVID-19 diagnosis.[26]

The validation of test kits requires time, putting additional strain in response to the rapid spread of the disease. Thus low sensitivity (30%) was reported in the RT-PCR testing kits in the early stage of the pandemic.[29] In a recent review conducted by Younes et al, RT-PCR testing kits developed in the US, France and Germany had a high sensitivity, at 95% but specificity was not reported.[26] Yan et al have suggested that human RNase P gene could be amplified as an internal control to reduce false-negative results and template volume could be increased to improve sensitivity.[23] One study from March 2020 in India found no false negatives but only 5 in 7 positive tests were confirmed.[20] As a consequence, all apparent positive tests were retested by the National Institute of Virology (NIV). Point of care testing would have many benefits in terms of speed of results but Döhla et al reported that available tests had low sensitivity (36.4%) compared to RT-PCR in a small group of 39 COVID-19 patients.[28] Given concerns about the limitations in sampling and testing, there is a need for caution about potential false negative results, and repeated testing may be required if symptoms persist.

Linked data to facilitate active case finding

There are several examples of how linked health service data can be used to identify cases.[16,30] For example, the digital COVID-19 boarder control system collect health information of inbound passengers. The data is linked to the "epidemic prevention tracking system" used by the health authorities for suspect case isolation and quarantine, and simultaneously linked to the National Health Insurance MediCloud System, a cloud-based platform for medical professionals to gain updated clinical data of the patient seeking care.[31] The data linkage

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provides medical staff with real-time information on the patient's COVID-19 travel and contact history, home isolation and quarantine status, to provide necessary care with appropriate infectious control measures.[32] Screening high-risk populations (such as health care workers) has been used for active case finding.[33]

2. Adherence to isolation and local support needs

Inbound traveller quarantine

Border controls and restricted entry have been applied in East Asian countries early during the pandemic.[29] These have been followed by comprehensive inbound traveller quarantine. For example, Singapore imposed a 14-day "Stay Home Notice" (SHN) on visitors and returning residents from COVID-19 endemic areas since January 2020 and for all inbound traveller from 9th April 2020. Upon arrival, travellers underwent a 14-day mandatory stay in government-designated hotels. During this period, they were not allowed to leave their individual rooms, where specific infectious disease prevention procedures were implemented.[34]

Timely identification of contacts

Ferretti et al. reported an estimated reproduction number of SARS-Cov2 of 2.0, of which 0.9 was infection occurred during the pre−symptomatic stage, suggesting about half cases were infected by COVID-19 patients before symptom onset.[35] An individual infected by SARS-Cov2 can actively shed the virus in their respiratory system 6 days before and 3 weeks after displaying symptoms, depending on disease severity.[23,26] While the virus detected in respiratory samples peaked within the first week of the illness, virus shedding increases in stool sample and peaked in week 2 or 3.[23] It thus is essential to stop the transmission by rapidly finding potential cases and their contacts and sufficiently quarantining. Countries may apply an additional week of self-health management beyond the 2-week quarantine period.[16]

Digital tools to facilitate FTTIS

Conventional epidemiological contact tracing, which relies on personal interviews is labour-intensive and time-consuming, and may be challenging during a large-scale epidemic.[16] Countries thus apply digital tools in COVID-19 response. Telecommunication provider-based measures are more efficient than voluntary digital tools (such as Apps). The level of public discussion about privacy issues varies in different countries, in part reflecting democratic processes.[36-38]

Types of telecommunication provider-based measures may include:

1. Mapping the amount of anonymized cell phone movement in a particular area (for example, Germany, Austria, Italy),
2. Base station triangulation to approximate cell phone location (Taiwan) or
3. Access to GPS data generated by the phone (Israel)

Voluntary provision of data includes:

1. An app recording mobile devices within a contact range via Bluetooth (Singapore, Austria),
2. An app recording daily symptoms (South Korea, Taiwan, Poland) or
3. A QR code for entry or exit in key locations (China).

Aslam et al summarised digital tools used in East Asian countries for COVID-19 response by their level of intrusiveness on users.[29] Low intrusive digital tools provide services to individuals and act as the interface between governmental management and information service users.[30] Such tools are widely accepted by the population voluntarily. Other tools are used to enforce governmental quarantine policy and may involve individual movement restrictions and privacy concerns. These are subject to debate and such tools have lower voluntary acceptability, yet large scale uptake (60% to 75% of the population) may be necessary for them to be effective.[37] For example, in Singapore, the initial voluntary use of the contact tracing App was only 12%.[37] In fully democratic states, the use of such Apps require a clear legal basis for the information which can be used exclusively for infectious disease control and comes with strong legal limits on data access.[29,38]

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Support for individuals complying with self-isolation

Webster et al. identified factors associated with adherence, which included the knowledge people had about the disease and quarantine procedures, social norms, perceived benefits of quarantine and perceived risk of the disease, as well as practical issues such as access to supplies or the financial consequences of being out of work.[39] Self-isolation or quarantine may not be affordable for low-wage and gig economy workers.[40] To address such financial needs, the Singaporean government provides self-employed persons and businesses with employees undergoing self-isolation or quarantine (SHN) a $100 per day support. Deliveries of food and other supplies can be arranged for individuals in SHN, through a designated hotline. By law, landlords and dormitory operators cannot evict tenants in SHN. For individuals in residences that may not be suitable for SHN, the government offers hotels as an alternative.[34]

Psychological stress,[39,41,42] cognitive dissonance[38] or lack of physical activities[43] during self-isolation may adversely affect mental or physical health. In a rapid review, Brooks et al reported individuals in quarantine experiencing confusion, anger, post-traumatic stress symptoms, for reasons such as fear of infection, frustration, boredom, and inadequate supplies.[39] Strategies to mitigate these adverse effects include timely and sufficient information to reduce uncertainty, minimising quarantine interval to no longer than necessary, and providing adequate supplies for individuals serving self-isolation.

Razai et al. suggested validated psychological screening tools, such as the UCLA Loneliness Scale and Patient Health Questionnaire-4 for anxiety and depression, for use in identifying individuals who need support. This can help identify those who may benefit from counselling, coaching via telephone or online video consultations.[42] Non-medical social prescribing, such as visual choirs, online theatre or performances, exercise or art classes can reduce boredom. Communication by social media may ease loneliness. Banskota et al reviewed smartphone Apps to assist older adults coping with self-isolation, designed for social networking, food and drinks delivery, medical consultation, health and fitness.[44] Meinert et al described an agile process to develop an App for older people and their families to improve well-being while observing social distancing rules.[45]

3. How would real-time data management, linkage of datasets, and dashboards be developed, and who would "own this"?

APP and GPS

In South Korea, the government links digital databases to facilitate contact tracing, including electronic health records, phone-based GPS, card transaction records, and closed-circuit television.[36] Yasaka et al have proposed the use of an App with three guides, requiring users to create checkpoints (e.g. public spaces or shops with a QR code that can be scanned to the app), check risk level, and report COVID-19 status.[46] Likewise, in China, a QR code-based app is used for limiting the movement of suspected COVID-19 patients and displays a green, amber, or red code that is required to be in or enter locations, but the digital surveillance architecture and data protection mechanisms remain unclear.[29,37,38] Consequently, public concerns about data protection and privacy have been raised in respect to both the Chinese and South Korean Apps.

The efficacy of mobile positioning data (although not possible for 2G phones) was studied in Nigeria by Ekong et al. based on systems used in South Korea, Singapore and China.[47] A legal framework for data protection was suggested when implementing these systems and a third-party agreement was put in place for data use.

Mobile geopositioning data

Mobile geopositioning has been used to study the mobility, disease connectivity, and health risks in travellers.[48] These have an accuracy of 150 meters so they reduce the risk of undermining individual privacy.[16]

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Taiwan has applied a geopositioning method to facilitate adherence to home isolation or quarantine, rapidly identifying contacts in a suspected large-scale outbreak. An example is the identification of 627,386 contacts of the 3,000 Diamond Princess passengers touring in Taiwan during COVID-19 outbreak.[16] The process took a day and text messages advising self-health management were sent to all contacts. Contact data were linked to electronic health records for follow-up and testing of symptomatic contacts. Geopositioning data has lessened the pressure on health authorities who would otherwise have relied upon resource-intensive manual contact tracing, facilitating timely large-scale outbreak containment. The resources saved by leveraging technology can then be used for taking care of vulnerable populations and those without access to a mobile phone.

4. How would a "rapid response" occur, and what would precipitate such a response?

On 22nd January, an individual from Wuhan who had travelled to Hong Kong reported respiratory symptoms, and a second suspected case was reported the next day. Both received medical care, being placed in isolation, and later tested positive for COVID-19. Contact tracing started immediately and the travel histories of patients were retrieved and published online. All their close contacts, including passengers seated close by and taxi drivers, were subject to quarantine at a holiday village converted to a quarantine centre to host contacts of confirmed cases. A hotline was set up to answer public enquiries, especially for passengers on the same train/flight.[45]

In Singapore, proactive contact tracing and cluster identification are key features of the country's COVID-19 response. Health professionals are trained to identify potential outbreak clusters, asking a series of questions to COVID-19 patients. The Ministry of Health works with hotels to develop quarantine sites and liaises with those possessing CCTV footage to track cases.[45]

Infection control

Inadequate infection control during isolation or quarantine may increase the transmission of SARS-Cov-2. Close quarter isolation was found to be ineffective [49] in the case of the Diamond Princess. The virus spread to 634 passengers despite contact tracing[50] and safety measures,[51] leading the authors to conclude that aerosol transmission played an important role in confined settings. However, this was questioned by Wang et al in a hospital study of transmission through air, sewage, surfaces and personal protective equipment in which swabs tested positive only for sewage with virus cultures not being found.[52]

Nosocomial infection and cross infection of health workers was reported at the Sheffield Teaching Hospitals NHS Foundation Trust, of 1,533 symptomatic healthcare workers, 18% were positive for SARS-CoV-2.[53] It was estimated that a third of staff had completed a shift while symptomatic, emphasising the need for regular and efficient testing for healthcare workers, who have a high risk of infection, for the protection of vulnerable patients and civilians.

To control nosocomial infection, Taiwan implemented nationwide enhanced Traffic Control Bundling (eTCB) in hospitals whereby infection was controlled with a combination of triage prior to hospitalisation, separation between risk zones, strict PPE use and hand disinfection checkpoints.[30,54] Risk zones were divided into contamination, intermediate and finally, clean. As droplet and fomite transmission have been observed for COVID-19 inside and outside hospitals, containment of nosocomial transmissions benefited with eTCB. This drew on research by Yen et al in which SARS infection among HCWs was transmitted to 2HCWs (0.03 cases per bed) in the eTCB hospital compared with 50 probable cases (0.13 cases per bed) in the control group.[55] This strategy was implemented across Taiwan on 21st May, 2003 and within two weeks the epidemic was under control.

5. How would an app be assimilated in light of the above?

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The Singaporean government has developed the app “TraceTogether”, recording other users who have been in proximity to a smartphone user via Bluetooth. After a user is found to be positive, individuals at risk are contacted directly.[29,38] The EU PEPP-PT coalition proposed a privacy-friendly contact tracing apps with the use of matching Bluetooth signals, based on the Singaporean “TraceTogether” app.[37] The Korean Ministry of the Interior and Safety has developed a mobile phone application named "self-quarantine safety protection" App that monitors the location of the quarantined user, informs health authorities, allows the user to report on their symptoms, and health officials can evaluate if a test is needed.[36]

6. What are the barriers to & enablers of being tested, reporting contacts & isolating as a result of being contacted?

Logistics

Testing: With the United States in mind, Parmet et al. encouraged the promotion of “free testing” in order to reach poorer communities and eradicate the infection.[40] Although the UK already has free testing for adults, there are challenges in communicating information on how to obtain a test, in what circumstances and when. Mark et al. reviewed the feasibility of a mobile community testing team in Scotland and reported a lack of guidance on infection control for testing in the community, thereby risking cross-transmission. Other barriers included shortage of staff and long travel times.[56]

Contact tracing Validity and reliability of information recorded by the app may not be accurate or precise due to technological limitations.[44] Klonowska et al summarised the high coverage needed for an app to be effective. This may be difficult to reach on a voluntary basis, for issues such as mobile storage data, operating systems, and battery power to support constant Bluetooth activation, the fact that young children and senior citizens may not carry or own a personal smart device, and the issue of individual willingness.[37] In the UK, only 47% of individuals who are 75 years or older use the internet.[57] It is essential that outbreak control teams cater to their needs and those of community volunteers, such as the NHS Volunteer Responders in the UK.[42]

Isolation Logistical challenges arise in providing food, sanitation, transport[58] and care for individuals living in the restricted zone, especially if the epidemic spreads to regional or national levels.[40]

Public awareness and Communication

The number of tests carried out during an emergency will depend on the public perception of the reliability of testing services and the effectiveness of communication of actions that can be taken. It is also necessary to ensure that kits and location of testing stations are accessible to disadvantaged groups who may not have access to a smart phone or basic supplies. Those from ethnic minorities should be made aware of the higher death rates statistically observed in members and encouraged to test early. Regular press conferences by the central outbreak control team, briefing on the progress of the pandemic, changes in policies, and correcting misinformation[30] can increase public awareness of the need of FTTIS and clarify rumours generated from knowledge gaps and uncertainties.[54]

The internet and social media have an important influence on behaviour in isolation, which has grown with increased internet use since the COVID-19 lockdown. Farooq et al. tracked the effects of social media, news websites and emails as well as the living situation on the individual-level, and the intention of self-isolation during the pandemic in a panel of 225 participants.[59] It was found that while frequent social media use contributed to information overload and cyberchondria, it increased propensity for self-isolation. McNeill et al similarly studied the effects of tweets and found that social media played a role in the motivation to conform to health measures.[60] To motivate members of the public to self-isolate in a healthy way, a combination of lowering perceived response costs and clear information about the severity of risks should be implemented.[59] It is also suggested that messages of severity of disease be disseminated via reliable official networks such as

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government outreach, news and journalism, while social media focuses on hopeful messages and recommended health measures.

Legal issues

In Singapore, the Infectious Diseases Act provides a legal basis to prosecute non-compliance, with consequence of fines and/or imprisonment.[34] Similarly, the Communicable Disease Control Act in Taiwan mandates the government to implement effective measures in COVID-19 controls including FTTIS.[30]

In the US, the Public Health Service Act empowers the Surgeon General (delegated to the Centers for Disease Control and Prevention (CDC)) to enforce quarantine and isolation and provide medical care for detained individuals. Court rulings establish that quarantine cannot be imposed in a way that is racially discriminatory, and governments are required to have a strong legal basis for imposed restrictions.[40]

Public cooperation

The effective control of the highly contagious COVID-19 epidemic relies on the collaboration of individuals to voluntarily provide their information and comply with preventive measures. The potential undesirable outcome in telling the truth, such as adverse immigration consequences for noncitizens, deportation of undocumented migrants or refusal of entry for passengers with a fever, can be a barrier for these individuals, who may be at high risk of infection, to seek care or provide accurate information.[40] Special protection of personal information is needed, with the goal of facilitating trust, prioritising care, and encouraging individuals to cooperate, to prevent further transmission of the disease.

Economical support

To ease the economic pressure and uncertainty during large-scale quarantine, governments have established different schemes to support the individuals and families. East Asian countries have provided support schemes for employers, employees and households.[29] In the US, the government passed the Families First Coronavirus Response Act, including paid sick leave and unemployment insurance.[55] In the UK, national job retention schemes and financial support schemes have been rolled out early to support individuals and businesses during the COVID-19 epidemic.[55]

Conclusion

We identified in our systematic review the core elements for an effective Find, Test, Trace, Isolate, Support (FTTIS) system to interrupt the spread of a novel infectious disease, where treatment or vaccination are yet available, such as the current COVID-19 pandemic. We reported optimal strategies in the literature to shorten case finding time, improve both accuracy and efficiency of the test, coordinate stakeholders and sectors involved in a FTTIS system, support individuals to follow FTTIS and appropriately use digital tools to facilitate the program. Results of the review may inform countries considering implementing these measures.

Contributorship

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Patient and Public Involvement

There is no patient and public involvement in the study.

Declarations of interest

The authors declare no conflict of interests.

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Data sharing

Data are available in a public, open access repository.

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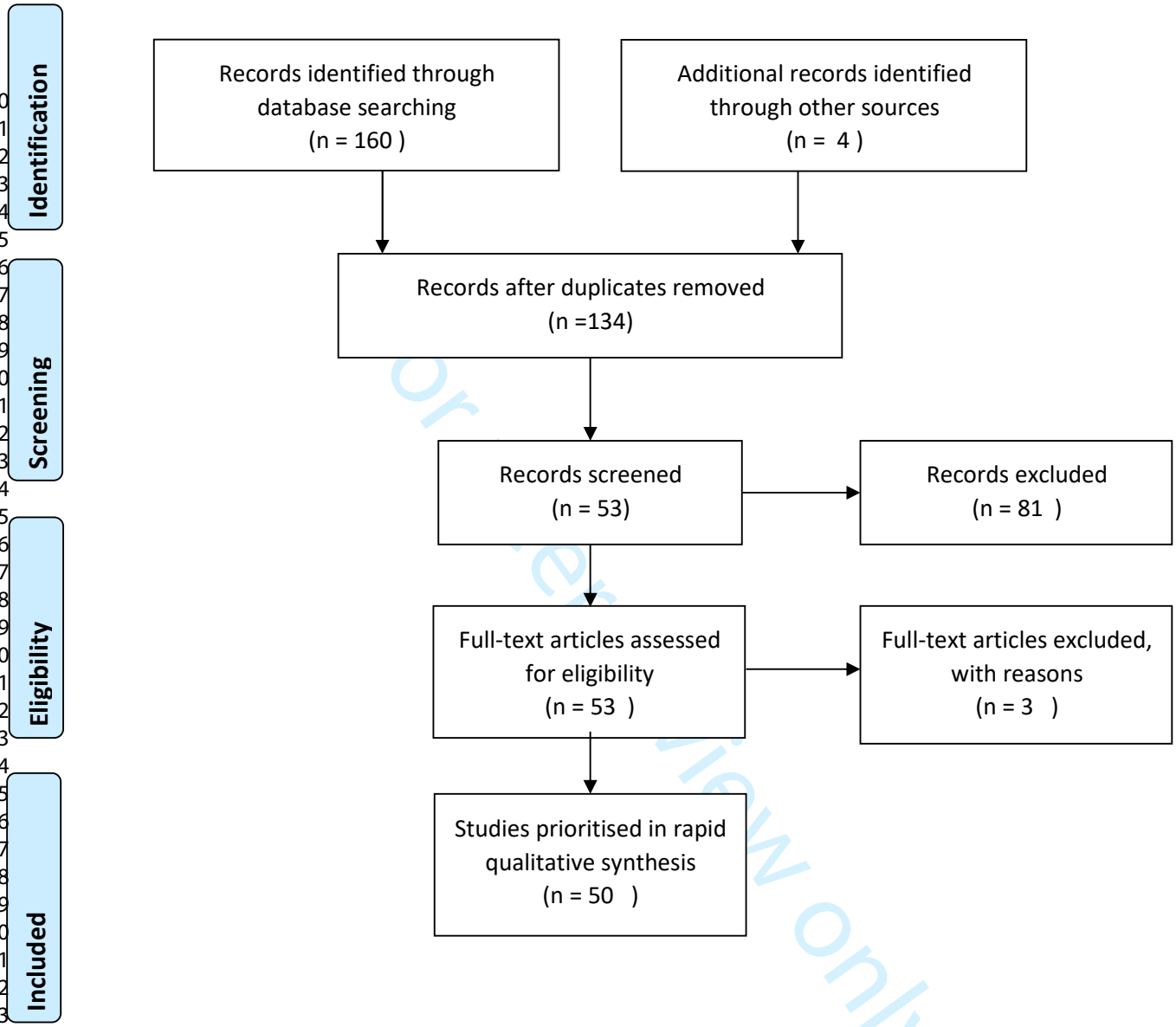
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Figure Legend

Figure 1: Systematic review study flow diagram

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Supplementary appendix for the study

A rapid systematic review and case study on find, test, trace, isolation and support policies for COVID-19 prevention and control

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Supplementary section one : List of questions to synthesise information from the eligible studies in the review

Study summary

Title

Author

Journal

Research question or objective

Study design

Study population

Sample size

Analysis or Method

Main findings

Implications

Main limitation(s)

Policy questions

1. How would a 24 hr turn-around from sampling to result actually work? (the report only goes as far as suggesting a 3 day turnaround)

2. What local governance and partnership structures are required, and what would a local “outbreak team” look like?

7. Adherence to isolation, and local support needs

3. What local commissioning arrangements are needed to facilitate this?

4. How would real-time data management, linkage of datasets, and dashboards be developed, and who would “own this”?

5. How would a “rapid response” occur, and what would precipitate such a response?- eg schools, care homes, other localities?

6. How would an app be assimilated in light of the above?

9. What are the barriers to & enablers of being tested, reporting contacts & isolating as a result of being contacted?

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Supplementary section two: List of reviewed articles:

Klonowska, Klaudia, and Pieter Bindt. The COVID-19 Pandemic: Two Waves of Technological Responses in the European Union. Hague Centre for Strategic Studies, 2020.

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Lessons from countries implementing find, test, trace, isolation and support policies in the rapid response of the COVID-19 pandemic: a systematic review

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Title: Lessons from countries implementing find, test, trace, isolation and support policies in the rapid response of the COVID-19 pandemic: a systematic review

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ABSTRACT

Objective To learn lessons from the experiences of countries implementing Find, Test, Trace, Isolate, Support (FTTIS) in the first wave of the COVID-19 pandemic.

Design, data sources and eligibility criteria We searched MEDLINE (PubMed), Cochrane Library, SCOPUS and JSTOR. Research articles and reviews on the use of contact tracing, testing, self-isolation and quarantine for COVID-19 management between were included in the first review.

Data extraction and synthesis We extracted information including study objective, design, methods, main findings and implications. These were tabulated and a narrative synthesis was undertaken given the diverse research designs, methods, and implications.

Results We identified and included 118 eligible studies in the review. We identified the core elements for an effective Find, Test, Trace, Isolate, Support (FTTIS) system needed to interrupt the spread of a novel infectious disease, where treatment or vaccination was not yet available, as with the initial stages of the COVID-19 pandemic. We report methods used to shorten case finding time, improve accuracy and efficiency of tests, coordinate stakeholders and sectors involved in a FTTIS system, support individuals isolating and make appropriate use of digital tools.

Conclusions We identified in our systematic review the effective strategies for a successful FTTIS system to interrupt the spread of a novel infectious disease. These include border controls, restricted entry, inbound traveller quarantine, and screening for case-finding; repeated testing to minimise false diagnoses and pooled testing in resource-limited circumstances; extended quarantine period and the use of digital tools for contact tracing and self-isolation. Support for mental or physical health and livelihood are needed for individuals undergoing self-isolation/quarantine. An integrated system with rolling-wave planning can best utilise effective FTTIS tools to respond to the fast-changing COVID-19 pandemic. Results of the review may inform countries considering implementing these measures.

Article summary

What is already known on this subject?

The initial COVID-19 prevention and control depend on effective case finding, quarantine and isolation.

Strengths and limitations of this study

- The study systematically reviews the core elements for an effective Find, Test, Trace, Isolate, Support (FTTIS) system to interrupt the spread of COVID-19. Results of the review may inform countries considering implementing these measures.
- We reported optimal strategies in the literature to shorten case finding time, improve both accuracy and efficiency of the test, coordinate stakeholders and sectors involved in a FTTIS system, support individuals to follow FTTIS and appropriately use digital tools to facilitate the program.
- The limitation of the study is the qualitative review, as there were too few quantitative studies in the subject area to enable a meta-analysis.

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INTRODUCTION

The initial interruption of the transmission of COVID-19 depends on the rapid isolation of infected individuals. For this to happen, a complex set of co-ordinated elements must be put in place to Find potential cases, either by identifying those who have symptoms or screening of individuals at risk, Test to confirm the presence of infection, Trace contacts, Isolate those infected and their contacts,[1] and Support[2] those in isolation to reduce the risk that they will breach any restrictions.

While various elements of a Find, Test, Trace, Isolate, Support (FTTIS) system have long been core elements of the public health role, COVID-19 has some specific characteristics, such as transmission from presymptomatic individuals[3] and a strong age gradient in both likelihood of infection and disease severity, as well as others that remain poorly understood. Despite this uncertainty, successful COVID-19 FTTIS programs have been reported to display coordination and smooth teamwork between the essential sectors, i.e. central and local government authorities, laboratories and the medical care system.[1] The sectors are interconnected by integrated and real-time information flow. However, methods to achieve this integral system remain inconclusive and largely determined by the existing organisations and infrastructures.

Digital tools are proposed to enhance the efficacy of FTTIS programmes.[1 3] Although the pilot study of the NHS contact tracing App showed positive results in the decrease of the rate of transmission,[4] challenges remain in its coverage and general usage for it to be effective.[5] Individuals or occupation groups with high COVID-19 risk will need additional support to access the programme[2] or reduce the burden of unnecessary or repeated episodes of quarantine. Other challenges for adopting a FTTIS programme include an optimal strategy to shorten time to find the case and contacts,[1] accuracy of the test,[1] establishing the integrated system,[6 7] barriers in adhering to FTTIS regulations,[8 9] and perceived economic impact on the livelihood of individuals.[9] To inform and support the design, implementation and betterment of a FTTIS system, we have undertaken a review and structured it to answer a series of key questions for a successful FTTIS during the urgent response of the COVID-19 pandemic. These policy questions included 1. Infrastructure and procedure of a sampling-to-results process that is 24 hours or less. 2. Adherence to isolation and local support needs. 3. How would real-time data management, linkage of datasets, and dashboards be developed, and who would “own this”? 4. How would a “rapid response” occur, and what would precipitate such a response? 5. How would an app be assimilated in light of the above? 6. What are the barriers to & enablers of being tested, reporting contacts & isolating as a result of being contacted?

METHODS

The systematic review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. The review protocol is described as the following.

Eligible studies are research articles and reviews evaluating the effectiveness of contact tracing, testing, self-isolation and quarantine on COVID-19 management published in English worldwide during the urgent response of the COVID-19 pandemic (December, 2019 to May, 2020 and June 2020 and January 2021).

Information sources and search, we searched MEDLINE (PubMed), Cochrane Library, SCOPUS and JSTOR. The search terms included "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title or text. The full search strategy is included in the supplementary appendix section one, and we apply the same search criteria. The date last searched is 28th May 2020. Studies were limited to those in which the subjects were human. Relevant reports and literature cited in the papers identified, as well as material from other sources such as the COVID Response Monitor developed by the European Observatory on Health Systems and Policies were also considered.[10] As research articles published after May 2020 may include additional information on FTTIS during the urgent

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response, we supplement our search with a second systematic review discussing FTTIS strategies during the early phase of the COVID-19 pandemic published between June 2020 and January 2021 (supplementary section two).

Study selection In the MEDLINE search, clinical studies, clinical trials, evaluation studies, government documents, journal articles, multicentre studies, observational studies, practice guidelines, pragmatic clinical trials and technical reports were studied. Reviews were included in the initial review. Studies whose subject area were in medicine or social sciences in SCOPUS, and research reports in JSTOR (excluding book chapters) were selected. We screened title, abstract and content to remove studies unrelated to FTTIS or COVID-19.

Data collection process and data items For each publication we extracted (where appropriate) information on study design, method, results, main findings, and limitations using the PICOS tool.[11] **Summary measures and synthesis of results** Following discussion with researchers involved in the COVID-19 response in the UK (https://www.independentsage.org/), a list of key questions to answer was developed and answered using the literature collected. **Risk of bias in individual studies** As studies eligible for policy review were mostly qualitative, we managed by using the Critical Appraisal Skills Programme (CASP) tool to assess the methodological rigour of included studies.[12]

Patient and Public Involvement The study questions were inspired by the discussion with the voluntary public scientific group for COVID-19 response in the UK (https://www.independentsage.org/).

RESULTS

For the initial search, we identified 50 studies from PUBMED (Medline), 43 studies from SCOPUS, 56 from JSTOR, 1 from the Cochrane library and 4 from additional sources. The flow diagram is set out in Figure 1. From the 164 initial publications, we removed 30 duplicate studies, 84 items that were not directly focused on COVID-19 or strategy of testing, contact tracing or isolation, or not in the format of a research report (such as editorials or podcasts). We subsequently identified another 58 studies from the second supplementary systematic review discussing FTTIS strategies during the early phase of the COVID-19 pandemic published between June 2020 and January 2021. We thus included a total of 118 studies in the review. We found eligible studies clearly stated aims, applied appropriate methods, although none addressed all proposed study questions, they provided data relevant to an aspect in our review.

Overview of FTTIS

To set the scene, we begin with a non-exhaustive review of some key findings on the COVID-19 pandemic. A review by Nussbaumer-Streit B et al. concluded that a combination of quarantine and other nonpharmaceutical interventions would significantly reduce COVID-19 incidence and mortality.[13] Policy decisions must be informed by a detailed understanding of transmission dynamics. With growing evidence that so-called super-spreading events play an important role in many outbreaks,[14] that finding, testing and isolating those present at such events is a priority.[15] This relates to work by Manchein et al who have proposed that the growth of COVID-19 cases follows a power-law pattern rather than an exponential one, suggesting that transmission is characterised by features of scale free networks, fractal kinetics, and small world features. The implication is that while social distancing has been important to slow the spread of COVID-19 so far, a robust testing and isolation will be necessary to facilitate economies to reopen.[16] These conclusions are consistent with work from Italy stressing the importance of maintaining very strict quarantine of those affected. [17] Policies must also be underpinned by high quality data, yet some countries have so far struggled to put well-functioning integrated systems in place. Inadequate testing will often underestimate the true number and growth rate of COVID-19 cases.[18] [19] This is a problem because testing is at the heart of a comprehensive FTTIS system. It is the only way to make a diagnosis where symptoms are atypical or unclear or in cases that are asymptomatic or presymptomatic.[20] For incidence, analysing surveillance data found that

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the high case-fatality during the first 40-day of COVID epidemic in Italy was primarily due to insufficient testing to accurately detect the outbreak, thus masked the transmission in the community and healthcare facilities.[21] In countries where FTTI interventions implemented subsequently, there may be a delay in their effect on disease control.[22] There are now a number of reports on how FTTIS systems have been designed and operated.[23]

1. Infrastructure and procedure of a sampling-to-results process that is 24 hours or less.

Countries have differed in whether they have built on existing infrastructure or created new structures to deal with the pandemic. We have identified two modalities from the reviewed studies, one is a patient-based and the other a laboratory network. The UK has initially largely adopted the former approach, and methods include drive-through SARS-Cov2 testing[24] with patients being informed of their results within 24 to 36 hours. Lagier et al reported on SARS-Cov2 RT-PCR testing of repatriated French citizens during their initial response,[25] and accelerated time from sample to result to 3 hours by prioritising extraction of RNA from samples. Replacing RNA extraction step by performing RT-PCR directly on heat-inactivated or lysed samples could accelerate the process further.[26] Point of care testing could also reduce the time to results to less than 2 hours.[27]

Binniker has set out the steps that should be taken when scaling up activity in existing laboratories,[28] which has the advantage of using existing networks, including transport and communications. Similarly, Gupta et al have described the response of laboratories in universities and clinics in India being converted to testing centres.[29] Tolia et al described the implementation of a testing programme within a hospital emergency department.[30]

Strategies to improve the testing response

The case definition of COVID-19 changed frequently in the initial stage of the epidemic, as knowledge of the disease evolved. In the early stage of the epidemic, test kits and laboratory capacity were in short supply in many countries.[31] To investigate the potential SARS-Cov-2 local transmission, the Dutch National Outbreak Management Team conducted a rapid screening among health care workers in an area of suspected COVID-19 outbreak. Testing followed a national protocol and was carried out either locally or in central laboratories. Nine hospitals were asked to sample health care workers on Saturday, 7th March and results were due on Monday, 9th March. When the SARS-Cov-2 positive rate came back as 4.1% among 1,097 tested, the local outbreak was confirmed.

Other laboratory-based strategies to improve testing efficiency include Yan et al. suggesting placing samples in reagents containing guanidine salts, for example, TRIZOL, TRIZOL LS, or AVL buffer, to inactivate the virus and protect RNA.[32] Pool-based testing approaches have been suggested to increase the testing capacity[33 34] and facilitate the detection of asymptomatic cases.[35] Only when the pooled batch tested positive, the samples contributing to the batch were tested individually.[36] A similar three-stage pooling system was reported by Eberhardt et al who showed that it was possible to test three and seven times as many individuals with the same number of test kits when prevalence rates were 5% and 1%, respectively.[37]

Modelling studies reported strategies testing low-risk versus high-risk symptomatic individuals was effective,[38 39] given it can be accurate and well-targeted.[40] A more targeted testing, measured by testing coverage (tests per confirmed case) was reported to be correlated with a greater reduction in COVID fatality than population testing number (tests per million people) among OECD countries and Taiwan.[41] An effective contact tracing and testing coverage may be indicated by a shorter interval detected between the diagnosis dates of the index case and secondary cases.[42 43]

Minimising false testing results.

False testing results can impede both the treatment of the patient and public health interventions to contain COVID-19 infection.[44] The testing accuracy may be influenced by sampling and testing kit performance. The

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gold standard for COVID-19 diagnosis remains RT-PCR–based assays with respiratory specimens.[45] Among different respiratory sampling methods, RT-PCR assays based on nasopharyngeal swabs tend to have a lower cycle threshold than oropharyngeal swabs, indicating a higher viral load collected through nasopharyngeal swabs.[32] Individuals who are initially RT-PCR negative can become positive with repeated tests.[46 47] The authors attribute this to differences in the amount of virus at the site samples and viral shedding at the time of sampling.[46-48] Sampling error can occur due to low levels of virus at the beginning or end period of shedding and inadequate swabs. Thus, it was advocated taking multiple samples,[46 48 49] and supporting RT-PCR results with Chest Computerized Tomography imaging for COVID-19 diagnosis.[45 50]

The validation of test kits requires time, putting additional strain in response to the rapid spread of the disease. Thus low sensitivity (30%) was reported in the RT-PCR testing kits in the early stage of the pandemic.[51] In a review by Younes et al, RT-PCR testing kits developed in the US, France and Germany had a high sensitivity, at 95% but specificity was not reported.[45] Yan et al have suggested that human RNase P gene could be amplified as an internal control to reduce false-negative results and template volume could be increased to improve sensitivity.[32] One study from March 2020 in India found no false negatives but only 5 in 7 positive tests were confirmed.[29] As a consequence, all apparent positive tests were retested by the National Institute of Virology (NIV). Point of care testing would have many benefits in terms of speed of results but Döhla et al reported that available tests had low sensitivity (36.4%) compared to RT-PCR in a small group of 39 COVID-19 patients.[48]

Besides RT-PCR testing, serologic antibody assays detect immunoglobin G and immunoglobin M to SARS-CoV-2 after infection.[52] However, due to the validity of the serologic testing and uncertainty in the duration of the immune response, results of serologic testing have been used to determine if an individual or the proportion of a population have previously infected.[53] With Given concerns about the limitations in sampling and testing, there is a need for caution about potential false negative results, and repeated testing may be required if symptoms persist. Testing at the end of the quarantine may help identify and care for delayed or asymptomatic COVID-19 cases while preventing further transmission.[54]

Linked data to facilitate active case finding

There are several examples of how linked health service data can be used to identify cases.[23 55] For example, the digital COVID-19 boarder control system collects the health information of inbound passengers. The data is linked to the "epidemic prevention tracking system" used by the health authorities for suspect case isolation and quarantine, and simultaneously linked to the National Health Insurance MediCloud System, a cloud-based platform for medical professionals to gain updated clinical data of the patient seeking care.[56] The data linkage provides medical staff with real-time information on the patient's COVID-19 travel and contact history, home isolation and quarantine status, to provide necessary care with appropriate infectious control measures.[57] Screening high-risk populations (such as health care workers) has been used for active case finding.[58]

2. Adherence to isolation and local support needs

Inbound traveller quarantine

Border controls and restricted entry have been applied in East Asian countries early during the pandemic.[51] These have been followed by comprehensive inbound traveller quarantine. For example, Singapore imposed a 14-day "Stay Home Notice" (SHN) on visitors and returning residents from COVID-19 endemic areas since January 2020 and for all inbound traveller from 9th April 2020. Upon arrival, travellers underwent a 14-day mandatory stay in government-designated hotels. During this period, they were not allowed to leave their individual rooms, where specific infectious disease prevention procedures were implemented.[59]

Timely identification of contacts

Ferretti et al. reported an estimated reproduction number of SARS-Cov2 of 2.0, of which 0.9 was infection occurred during the pre−symptomatic stage, suggesting about half cases were infected by COVID-19 patients before symptom onset.[60] An individual infected by SARS-Cov2 can actively shed the virus in their respiratory system 6 days before and 3 weeks after displaying symptoms, depending on disease severity.[32 45] While the

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virus detected in respiratory samples peaked within the first week of the illness, virus shedding increases in stool samples and peaked in week 2 or 3.[32] It thus is essential to stop the transmission by rapidly finding potential cases and their contacts and sufficiently quarantining.[61] Modelling study reported minimising testing delay effectively reduce onward transmissions.[62] Countries may apply an additional week of self-health management beyond the 2-week quarantine period.[23]

Modelling studies reported strategies to reduce random community contacts preferable to reducing personal contacts,[38 63] quarantine days can vary depending on the reduction of contacts reached by individuals,[63] and testing on the beginning and the end of quarantine.[64] The definition of contact may vary, Korea applied a focused testing strategy that when a COVID-19 outbreak with an unknown source of infection occurred, testing was done on individuals who were working, living, or visiting in the outbreak area.[65] Individuals who tested positive were hospitalised, whereas individuals who tested negative were self-quarantined for 14 days. The strategy effectively reduced the proportion of COVID-19 cases with an unknown source of infection and the risk of transmission in the community.[65] The direction of contact tracing may be forward or both forward and backward, the later identifies unascertained or asymptomatic carriers and enable the isolation of otherwise undiscovered transmission.[27]

Digital tools to facilitate FTTIS

Conventional epidemiological contact tracing, which relies on personal interviews is labour-intensive and time-consuming, and may be challenging during a large-scale epidemic.[23 66] Countries thus apply digital tools in COVID-19 response. Telecommunication provider-based measures are more efficient than voluntary digital tools (such as Apps). The level of public discussion about privacy issues varies in different countries, in part reflecting democratic processes.[67-69]

Types of telecommunication provider-based measures may include:

1. Mapping the amount of anonymized cell phone movement in a particular area (for example, Germany, Austria, Italy),
2. Base station triangulation to approximate cell phone location (Taiwan) or
3. Access to GPS data generated by the phone (Israel)

Voluntary provision of data includes:

1. An app recording mobile devices within a contact range via Bluetooth (Singapore, Austria),
2. An app recording daily symptoms (South Korea, Taiwan, Poland) or
3. A QR code for entry or exit in key locations (China).

Aslam et al summarised digital tools used in East Asian countries for COVID-19 response by their level of intrusiveness on users.[51] Low intrusive digital tools provide services to individuals and act as the interface between governmental management and information service users.[55] Such tools are widely accepted by the population voluntarily. Other tools are used to enforce governmental quarantine policy and may involve individual movement restrictions and privacy concerns.[70] These are subject to debate and such tools have lower voluntary acceptability, yet large scale uptake (60% to 75% of the population) may be necessary for them to be effective.[68] For example, in Singapore, the initial voluntary use of the contact tracing App was only 12%.[68] In fully democratic states, the use of such Apps require a clear legal basis for the information which can be used exclusively for infectious disease control and comes with strong legal limits on data access.[51 69]

Support for individuals complying with self-isolation

Webster et al. identified factors associated with adherence, which included the knowledge people had about the disease and quarantine procedures, social norms, perceived benefits of quarantine and perceived risk of the disease, as well as practical issues such as access to supplies or the financial consequences of being out of

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work.[71] Self-isolation or quarantine may not be affordable for low-wage and gig economy workers.[72] To address such financial needs, the Singaporean government provides self-employed persons and businesses with employees undergoing self-isolation or quarantine (SHN) a $100 per day support. Deliveries of food and other supplies can be arranged for individuals in SHN, through a designated hotline. By law, landlords and dormitory operators cannot evict tenants in SHN. For individuals in residences that may not be suitable for SHN, the government offers hotels as an alternative.[59]

Psychological stress,[71 73 74] cognitive dissonance[69] or lack of physical activities[75] during self-isolation may adversely affect mental or physical health. While anxiety was reported to positively associate with an interest in security-related actions, perceived reactance related to a negative association and loneliness to nonaction.[76] In a rapid review, Brooks et al reported individuals in quarantine experiencing confusion, anger, post-traumatic stress symptoms, for reasons such as fear of infection, frustration, boredom, and inadequate supplies.[71] Austria and Germany reported an increase trend of domestic violence associated with COVID social restrictions.[77] Strategies to mitigate these adverse effects include timely and sufficient information to reduce uncertainty, minimising quarantine interval to no longer than necessary, and providing adequate supplies for individuals serving self-isolation.

Razai et al. suggested validated psychological screening tools, such as the UCLA Loneliness Scale and Patient Health Questionnaire-4 for anxiety and depression, for use in identifying individuals who need support. This can help identify those who may benefit from counselling, coaching via telephone or online video consultations.[74] Non-medical social prescribing, such as visual choirs, online theatre or performances, exercise or art classes can reduce boredom. Communication by social media may ease loneliness. Banskota et al reviewed smartphone Apps to assist older adults in coping with self-isolation, designed for social networking, food and drinks delivery, medical consultation, health and fitness.[78] Meinert et al described an agile process to develop an App for older people and their families to improve well-being while observing social distancing rules.[79]

3. How would real-time data management, linkage of datasets, and dashboards be developed, and who would "own this"?

In Austria, the nationwide online Covid-19 dashboard with detailed information in all districts has facilitated effective early response to COVID pandemic.[77] In South Korea, the government links digital databases to facilitate contact tracing, including electronic health records, phone-based GPS, card transaction records, and closed-circuit television.[67] Yasaka et al have proposed the use of an App with three guides, requiring users to create checkpoints (e.g. public spaces or shops with a QR code that can be scanned to the app), check risk level, and report COVID-19 status.[80] Likewise, in China, a QR code-based app is used for limiting the movement of suspected COVID-19 patients and displays a green, amber, or red code that is required to be in or enter locations, but the digital surveillance architecture and data protection mechanisms remain unclear.[51 68 69] Consequently, public concerns about data protection and privacy have been raised in respect to both the Chinese and South Korean Apps.

The efficacy of mobile positioning data (although not possible for 2G phones) was studied in Nigeria by Ekong et al. based on systems used in South Korea, Singapore and China.[81] A legal framework for data protection was suggested when implementing these systems and a third-party agreement was put in place for data use.

Mobile geopositioning data

Mobile geopositioning has been used to study the mobility, disease connectivity, and health risks in travellers.[82] These have an accuracy of 150 meters so they reduce the risk of undermining individual privacy.[23] Taiwan has applied a geopositioning method to facilitate adherence to home isolation or quarantine, rapidly identifying contacts in a suspected large-scale outbreak. An example is the identification of 627,386 contacts

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of the 3,000 Diamond Princess passengers touring in Taiwan during COVID-19 outbreak.[23] The process took a day and text messages advising self-health management were sent to all contacts. Contact data were linked to electronic health records for follow-up and testing of symptomatic contacts. Geopositioning data has lessened the pressure on health authorities who would otherwise have relied upon resource-intensive manual contact tracing, facilitating timely large-scale outbreak containment. The resources saved by leveraging technology can then be used for taking care of vulnerable populations and those without access to a mobile phone.

4. How would a "rapid response" occur, and what would precipitate such a response?

On 22nd January, an individual from Wuhan who had travelled to Hong Kong reported respiratory symptoms, and a second suspected case was reported the next day. Both received medical care, being placed in isolation, and later tested positive for COVID-19. Contact tracing started immediately and the travel histories of patients were retrieved and published online. All their close contacts, including passengers seated close by and taxi drivers, were subject to quarantine at a holiday village converted to a quarantine centre to host contacts of confirmed cases. A hotline was set up to answer public enquiries, especially for passengers on the same train/flight.[79]

In Singapore, proactive contact tracing and cluster identification are key features of the country's COVID-19 response. Health professionals are trained to identify potential outbreak clusters, asking a series of questions to COVID-19 patients. The Ministry of Health works with hotels to develop quarantine sites and liaises with those possessing CCTV footage to track cases.[79]

Infection control

Inadequate infection control during isolation or quarantine may increase the transmission of SARS-Cov-2. Close quarter isolation was found to be ineffective [83] in the case of the Diamond Princess. The virus spread to 634 passengers despite contact tracing[84] and safety measures,[85] leading the authors to conclude that aerosol transmission played an important role in confined settings. However, this was questioned by Wang et al in a hospital study of transmission through air, sewage, surfaces and personal protective equipment in which swabs tested positive only for sewage with virus cultures not being found.[86]

Nosocomial infection and cross infection of health workers were reported at the Sheffield Teaching Hospitals NHS Foundation Trust, of 1,533 symptomatic healthcare workers, 18% were positive for SARS-CoV-2.[87] It was estimated that a third of staff had completed a shift while symptomatic, emphasising the need for regular and efficient testing for healthcare workers, who have a high risk of infection, for the protection of vulnerable patients and civilians. A study in Taiwan on the risks of contact infection before and after symptom onset showed that the spread of COVID-19 was highest within the first 4 days of infection. In comparison with transmission to medical staff, it is found that most transmission takes place in the family or social settings, partially due to the use of PPE and partially due to contact in the early stages of infection (pre-symptomatic).[88]

To control nosocomial infection, Taiwan implemented nationwide enhanced Traffic Control Bundling (eTCB) in hospitals whereby infection was controlled with a combination of triage prior to hospitalisation, separation between risk zones, strict PPE use and hand disinfection checkpoints.[55] [89] Risk zones were divided into contamination, intermediate and finally, clean. As droplet and fomite transmission have been observed for COVID-19 inside and outside hospitals, containment of nosocomial transmissions benefited with eTCB. This drew on research by Yen et al in which SARS infection among HCWs was transmitted to 2HCWs (0.03 cases per bed) in the eTCB hospital compared with 50 probable cases (0.13 cases per bed) in the control group.[90] This strategy was implemented across Taiwan on 21st May, 2003 and within two weeks the epidemic was under control.

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5. How would an app be assimilated in light of the above?

The Singaporean government has developed the app "TraceTogether" to support manual contact tracing,[91] recording other users who have been in proximity to a smartphone user via Bluetooth. After a user is found to be positive, individuals at risk are contacted directly.[51 69] The EU PEPP-PT coalition proposed a privacy-friendly contact tracing apps with the use of matching Bluetooth signals, based on the Singaporean "TraceTogether" app.[68] The Korean Ministry of the Interior and Safety has developed a mobile phone application named "self-quarantine safety protection" App that monitors the location of the quarantined user, informs health authorities, allows the user to report on their symptoms, and health officials can evaluate if a test is needed.[67] TRACE, the national contact tracing program in Taiwan, developed in 2017 has an integrated system to monitor health, trace contacts and link to other databases for a range of infectious diseases. The location of individuals could be tracked and communication between the user, Taiwan CDC, police and authorities as well as local health departments could be centralised through this platform for reporting symptoms and contact tracing.[92]

6. What are the barriers to & enablers of being tested, reporting contacts & isolating as a result of being contacted?

Logistics

The lack of public health funding may weaken the outbreak preparedness.[93] Regionalisation of the healthcare system might result in fragmentation in epidemic response and timely data availability.[21] healthcare resource is scarce, major barriers to COVID-19 control are the availability of and disparity to healthcare and testing.[94]

Testing:

With the United States in mind, Parmet et al. encouraged the promotion of "free testing" in order to reach poorer communities and eradicate the infection.[72] Although the UK already has free testing for adults, there are challenges in communicating information on how to obtain a test, in what circumstances and when. Mark et al. reviewed the feasibility of a mobile community testing team in Scotland and reported a lack of guidance on infection control for testing in the community, thereby risking cross-transmission. Other barriers included the shortage of staff,[95] shortage of testing kits and strict testing eligibility which excludes asymptomatic patients from receiving testing in time.[77]

Contact tracing Validity and reliability of information recorded by the app may not be accurate or precise due to technological limitations.[78] Klonowska et al summarised the high coverage needed for an app to be effective. This may be difficult to reach on a voluntary basis, for issues such as mobile storage data, operating systems, and battery power to support constant Bluetooth activation, the fact that young children and senior citizens may not carry or own a personal smart device, and the issue of individual willingness.[68] In the UK, only 47% of individuals who are 75 years or older use the internet.[96] Senior citizens in the Netherlands were also less likely to adopt a contact tracing App than their younger peers.[97] It is essential that outbreak control teams cater to their needs and those of community volunteers, such as the NHS Volunteer Responders in the UK.[74]

Isolation Logistical challenges arise in providing food, sanitation, transport[98] and care for individuals living in the restricted zone, especially if the epidemic spreads to regional or national levels.[72]

Public awareness and Communication

The number of tests carried out during an emergency will depend on the public perception of the reliability of testing services and the effectiveness of communication of actions that can be taken. It is also necessary to ensure that kits and the location of testing stations are accessible to disadvantaged groups who may not have access to a smart phone or basic supplies. Those from ethnic minorities should be made aware of the higher

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death rates statistically observed in members and encouraged to test early. Regular press conferences by the central outbreak control team, briefing on the progress of the pandemic, changes in policies, and correcting misinformation[55] can increase public awareness of the need for FTTIS and clarify rumours generated from knowledge gaps and uncertainties.[89] However, underreporting of COVID deaths by the government may occur in countries with incomplete vital registration or under political influence.[19 99]

The internet and social media have an important influence on behaviour in isolation,[100] which has grown with increased internet use since the COVID-19 lockdown. Farooq et al. tracked the effects of social media, news websites and emails as well as the living situation on the individual-level, and the intention of self-isolation during the pandemic in a panel of 225 participants.[101] It was found that while frequent social media use contributed to information overload and cyberchondria, it increased propensity for self-isolation. McNeill et al similarly studied the effects of tweets and found that social media played a role in the motivation to conform to health measures.[102] Based on populational surveys, Kowalski et al reported that conspiracy beliefs of COVID-19 were negatively associated with adherence to self-isolation guidelines in Poland.[103] To motivate members of the public to self-isolate in a healthy way, a combination of lowering perceived response costs and clear information about the severity of risks should be implemented.[101 104] It is also suggested that messages of the severity of disease be disseminated via reliable official networks such as government outreach, news and journalism, while social media focuses on hopeful messages and recommended health measures.

Legal issues

In Singapore, the Infectious Diseases Act provides a legal basis to prosecute non-compliance, with the consequence of fines and/or imprisonment.[59] Similarly, the Communicable Disease Control Act in Taiwan mandates the government to implement effective measures in COVID-19 controls including FTTIS.[55]

In the US, the Public Health Service Act empowers the Surgeon General (delegated to the Centers for Disease Control and Prevention (CDC)) to enforce quarantine and isolation and provide medical care for detained individuals. Court rulings establish that quarantine cannot be imposed in a way that is racially discriminatory, and governments are required to have a strong legal basis for imposed restrictions.[72]

Public cooperation

The effective control of the highly contagious COVID-19 epidemic relies on the collaboration of individuals to voluntarily provide their information and comply with preventive measures. In low or middle income countries, low public trust due to government corruption impedes public cooperation to observe FTTI regulations.[105] The potential undesirable outcome in telling the truth, such as adverse immigration consequences for noncitizens, the deportation of undocumented migrants or refusal of entry for passengers with a fever, can be a barrier for these individuals, who may be at high risk of infection, to seek care or provide accurate information.[72] Results from a multinational survey in France, Germany, Italy, the UK and US showed high public acceptability of a contact-tracing app.[106] Perceived benefits, self-efficacy[107], and trust[108] were positively associated with contact tracing apps uptake. Special protection of personal information is needed, with the goal of facilitating trust, prioritising care, and encouraging individuals to cooperate, to prevent further transmission of the disease.

Economical support

To ease the economic pressure and uncertainty during large-scale quarantine, governments have established different schemes to support individuals and families. Income compensation during quarantine has shown to double the compliance proportion of self-isolation.[109] East Asian countries have provided support schemes for employers, employees and households.[51] In the US, the government passed the Families First Coronavirus Response Act, including paid sick leave and unemployment insurance.[90] Countries implemented funding,

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financial stimulus, loans, aid packages to support the economy[77] or individual livelihood.[110] In the UK, national job retention schemes and financial support schemes have been rolled out early to support individuals and businesses during the COVID-19 epidemic.[90]

Discussion

We identified in our systematic review the core elements for an effective Find, Test, Trace, Isolate, Support (FTTIS) system to interrupt the spread of a novel infectious disease, such as the COVID-19 pandemic. The optimal strategies for a successful FTTIS during COVID-19 rapid response include:

Finding Border controls, restricted entry, and inbound traveller quarantine applied early during the pandemic have both reduced the spread of COVID-19 and facilitated case-finding. Screening high-risk populations (such as health care workers or inhabitants of long-term care facilities) is used for active case finding, especially during the outbreak investigation.

Testing During the urgent response of the pandemic, re-purpose existing laboratory network for testing is the preferred strategy for a rapid response, while establishing new testing sites to meet the needs as the pandemic progressed. In emergency or resources-limited circumstances, pooled testing can be used to improve testing efficiency. If symptoms persist or recur in individuals, repeated testing may be necessary to minimise false negative results from sampling error due to low levels of virus at different phases of shedding and inadequate swabs.

Trace and Isolate Digital tools such as apps, GPS, and mobile geopositioning data are deployed by countries for contact tracing and assess the compliance of self-isolation. As an individual infected by SARS-Cov2 can actively shed the virus 3 weeks or longer after symptoms onset, the duration of quarantine may vary. A two-week quarantine period with extension when required has been adopted by various countries.

Support can be in the form of mental health or physical health promotion or monetary aid, such as timely and adequate information to reduce uncertainty and anxiety, providing adequate supplies or allowance for individuals. These support also facilitate public trust that is fundamental in the joint COVID-19 response in the population, as it relies on individuals to voluntarily provide their information and adhere to disease control regulations.

Integrated system Although these measures, often combined, are implemented in different countries, the level of preparedness, timely implementation and scale up of effective measures made the difference between [111] or within[112] countries. For incidence, insufficient testing and quarantine led to the undetected wide transmission of SARS-Cov-2, resulting in high case-fatality during the first month of COVID epidemic in Italy,[21] and similarly in the Middle East[19] or during the first wave in the UK.[113]

A successful FTTIS requires seamless integration of all the components and the ability to integrate vulnerable populations or ethnic minorities. Linked health service data can contribute to the cooperation and communications among responsible authorities (central and local government, borders, laboratories and the medical care system) through real-time data sharing and dashboards. **Public collaboration** In additional to support for individuals, regular press conferences by the central outbreak control team to provide a briefing on the progress of the pandemic, changes in policies, and correcting misinformation can increase public awareness of the need for FTTIS. Open and balanced discussions on public concerns, such as personal data privacy and protection and limitations on individuals freedom for the public good, are essential.

Agile process The uncertainty and rapid progression of the pandemic make it important for policy and governance to have the flexibility to quickly adapt to the changes and challenges posed by the epidemic. A rolling-wave planning can facilitate authorities to learn, apply and refine tools for COVID-19 prevention and control, and the agile process allows better response in emergencies.

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Strength We found no published systematic review thus far addressing simultaneously the finding, test, tracing, isolation and support in the early response of COVID-19. Results of our review thus may inform countries considering implementing these measures. Previous reviews addressing each of the FTTIS components were summarised in our results. **Limitations** most studies in our review were qualitative research, as there were too few quantitative studies in the subject area to enable a meta-analysis. We reviewed studies published in English thus enable the review team to cross-validate the results. Although information documented in other languages might be missed, our search included studies from all regions of the world and findings from the policy review might be robust to the limitation. Our study focused on the FTTIS in the early phase of the pandemic. The FTTIS beyond rapid response, though not within the scope of the present study, may be warranted for further research.

Conclusion

We identified in our systematic review the effective strategies for a successful FTTIS system to interrupt the spread of a novel infectious disease. These include border controls, restricted entry, inbound traveller quarantine, and screening for case-finding; repeated testing to minimise false diagnoses and pooled testing in resource-limited circumstances; extended quarantine period and the use of digital tools for contact tracing and self-isolation. Support for mental or physical health and livelihood are needed for individuals undergoing self-isolation/quarantine. An integrated system with rolling-wave planning can best utilise effective FTTIS tools to respond to the fast-changing COVID-19 pandemic. Results of the review may inform countries considering implementing these measures.

Contributorship

All authors contributed to the study design, data collection, analysis, or results interpretation (SCC, SM, NT, AA, IA, S-LY, KK, MM, SM, DP). SCC, SM, NT accessed, analysed the data in the study and wrote the report. SCC had final responsibility for the decision to submit for publication.. SCC, SM, NT accessed, analysed the data in the study and wrote the report. SCC had final responsibility for the decision to submit for publication.

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There is no patient and public involvement in the study.

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The authors declare no conflict of interests.

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Data sharing

Data are available in a public, open access repository.

Ethical Approval Statement

Ethical Approval Statement is not applicable in the study.

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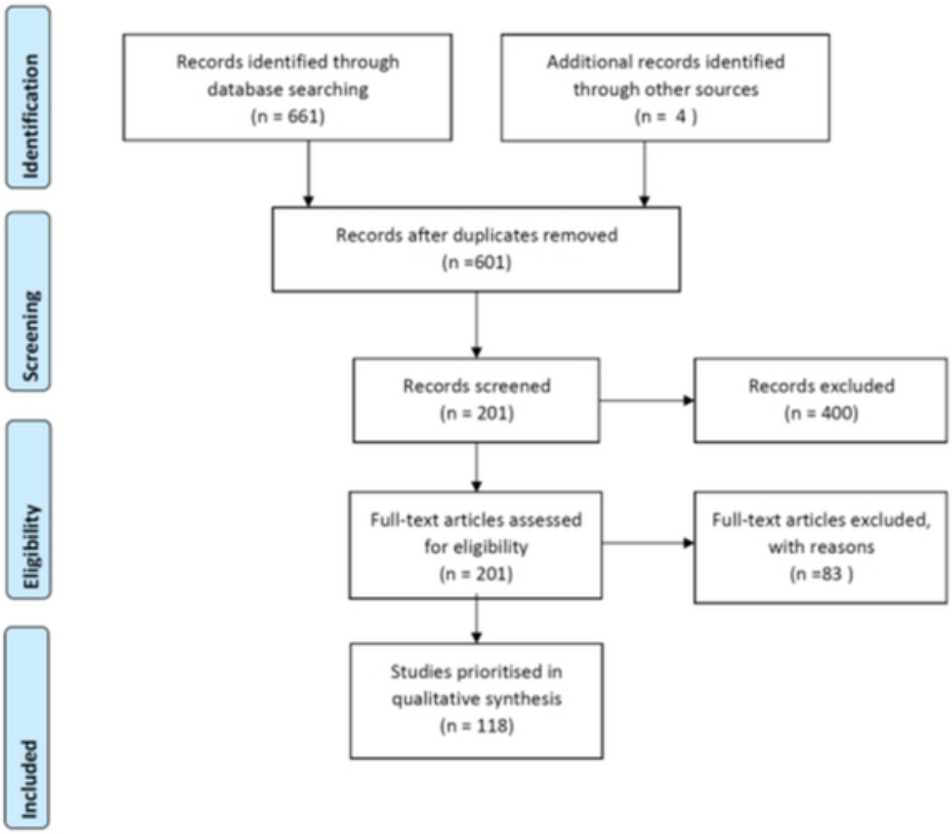
Figure Legend

Figure 1: Systematic review study flow diagram

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Figure 1: systematic review study flow diagram



Figure

50x47mm (300 x 300 DPI)

Supplementary appendix for the study

Lessons from countries implementing find, test, trace, isolation and support policies in the rapid response of the COVID-19 pandemic: a systematic review

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Supplementary section one: Search strategy

Excample database: MEDLINE (PubMed)

We applied search terms "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title or text. The studies were published between May 31th 2019 and May 31th 2020.

MEDLINE (PubMed) search: "contact"[Title] OR "contact tracing"[Title] OR "testing"[Title] OR "isolation"[Title] OR "quarantine"[Title] OR "self-isolation"[Title]) AND ("Covid-19"[Title/Abstract] OR "COVID-19"[Title/Abstract] OR "coronavirus"[Title/Abstract])

Filters applied: Clinical Study, Clinical Trial, Controlled Clinical Trial, Evaluation Study, Government Document, Journal Article, Meta-Analysis, Multicenter Study, Observational Study, Practice Guideline, Pragmatic Clinical Trial, Randomized Controlled Trial, Review, Systematic Reviews, Technical Report, in the last 1 year, Humans, English, MEDLINE.

We applied the same search strategy for Cochrane Library, SCOPUS and JSTOR.

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Supplementary section two: supplementary search strategy for second review between June 2020 and January 2021

We applied search terms "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title. The studies were published between June 1st 2020 and January 21st 2021.

Excample database: MEDLINE (PubMed)

("contact tracing"[Title] OR "testing"[Title] OR "self-isolation"[Title]) AND ("Covid-19"[Title] OR "COVID-19"[Title]) Filters: Clinical Trial, Controlled Clinical Trial, Evaluation Study, Government Publication, Journal Article, Meta-Analysis, Observational Study, Practice Guideline, Pragmatic Clinical Trial, Randomized Controlled Trial, Review, Systematic Review, Technical Report, Humans, English, MEDLINE, from 2020/6/1 - 2021/1/21

Result in 319 studies

Other exclusion criteria:

Items that were not directly focused on COVID-19 or strategy of testing, contact tracing or isolation in the general population, or not in the format of a research report, and studies with very small sample size <=5

After title screening: remove 180, 139 remains

After abstract screening: remove 49, 95 remains

We applied the same search strategy for Cochrane Library, SCOPUS and JSTOR.

The total studies after abstract screening: 108

The total studies after text screening: 58

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	3,4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	4
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	5
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	5
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

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Lessons from countries implementing find, test, trace, isolation and support policies in the rapid response of the COVID-19 pandemic: a systematic review

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Title: Lessons from countries implementing find, test, trace, isolation and support policies in the rapid response of the COVID-19 pandemic: a systematic review

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ABSTRACT

Objective To systematically learn lessons from the experiences of countries implementing Find, Test, Trace, Isolate, Support (FTTIS) in the first wave of the COVID-19 pandemic.

Design, data sources and eligibility criteria We searched MEDLINE (PubMed), Cochrane Library, SCOPUS and JSTOR, initially between 31th May 2019 and 21st January 2021. Research articles and reviews on the use of contact tracing, testing, self-isolation and quarantine for COVID-19 management were included in the first review.

Data extraction and synthesis We extracted information including study objective, design, methods, main findings and implications. These were tabulated and a narrative synthesis was undertaken given the diverse research designs, methods, and implications.

Results We identified and included 118 eligible studies. We identified the core elements of an effective Find, Test, Trace, Isolate, Support (FTTIS) system needed to interrupt the spread of a novel infectious disease, where treatment or vaccination was not yet available, as pertained in the initial stages of the COVID-19 pandemic. We report methods used to shorten case finding time, improve accuracy and efficiency of tests, coordinate stakeholders and actors involved in an FTTIS system, support individuals isolating and make appropriate use of digital tools.

Conclusions We identified in our systematic review the core elements of an FTTIS system needed to interrupt the spread of a novel infectious disease. These include border controls, restricted entry, inbound traveller quarantine, and comprehensive case-finding; repeated testing to minimise false diagnoses and pooled testing in resource-limited circumstances; extended quarantine period and the use of digital tools for contact tracing and self-isolation. Support for mental or physical health and livelihoods are needed for individuals undergoing self-isolation/quarantine. An integrated system with rolling-wave planning can best utilise effective FTTIS tools to respond to the fast-changing COVID-19 pandemic. Results of the review may inform countries considering implementing these measures.

Article summary

Strengths and limitations of this study

- The study systematically reviews the core elements of an effective Find, Test, Trace, Isolate, Support (FTTIS) system to interrupt the spread of COVID-19. Our findings can inform policy in future pandemics.
- We reported optimal strategies reported in the literature to shorten case finding time, improve both accuracy and efficiency of tests, coordinate stakeholders and actors involved in an FTTIS system, support individuals as they progress from case finding to isolation, and make appropriate use of digital tools to facilitate the program.
- The main limitation is the scarcity of quantitative studies.

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INTRODUCTION

Interrupting transmission of COVID-19 has depended upon rapid isolation of infected individuals. For this to happen, a complex set of co-ordinated elements must be implemented to Find potential cases, either by identifying those who have symptoms or examination of individuals at risk, Test to confirm the presence of infection, Trace contacts, Isolate those infected and their contacts,[1] and Support[2] those in isolation to reduce the risk that they will breach any restrictions.

While various elements of a Find, Test, Trace, Isolate, Support (FTTIS) system have long been core elements of the public health role, COVID-19 infection has some specific characteristics, such as transmission by pre-symptomatic and asymptomatic individuals[3] and a strong age gradient in disease severity, as well as other features that remain poorly understood. Despite this uncertainty, some countries have implemented successful COVID-19 FTTIS programs which have achieved coordination among relevant actors, including central and local government authorities, laboratories, and the medical care system.[1] These actors are interconnected by integrated real-time information flow. However, the ways that they have created these systems have varied, largely determined by existing structures.

There are no simple solutions. Thus, digital tools were initially seen as making a major contribution to FTTIS programmes.[1 3] Yet while the English NHS contact tracing App showed positive results in a pilot study, decreasing rates of transmission,[4] it struggled to achieve a sufficient level of uptake to be effective.[5] Individuals, organisations or occupational groups with high COVID-19 risk need additional support to use such apps[2] while it can be challenging to reduce the burden of unnecessary or repeated episodes of quarantine.

The main challenges in implementing an FTTIS programme include developing a strategy that can reduce delays in finding cases and contacts,[1] ensuring the accuracy of tests,[1] establishing integrated systems,[6 7] reducing barriers to adhering to regulations, especially isolating,[8 9] and mitigating adverse economic impacts on the livelihood of individuals affected.[9] To inform and support the design, implementation, and continuing improvement of an FTTIS system, we conducted a systematic review, structured to answer a series of key questions that arose during the initial response to the COVID-19 pandemic. These policy questions addressed the following issues: 1. Infrastructure and processes needed to achieve a sampling-to-results process taking 24 hours or less. 2. Adherence to isolation and local support needs. 3. Real-time data management, linkage of datasets, and dashboards, and ownership of the data generated. 4. Achieving a “rapid response”. 5. Assimilation of an application (app) in light of the above. 6. Overcoming the barriers to and enablers of being tested, reporting contacts & isolating as a result of being contacted.

METHODS

The systematic review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. The review protocol is described as the following.

Eligible studies are research articles and reviews evaluating the strategies of contact tracing, testing, self-isolation and quarantine on COVID-19 management published in English worldwide during the urgent response of the COVID-19 pandemic (May, 2019 to May, 2020 and June 2020 and January 2021).

Information sources and search, we searched MEDLINE (PubMed), Cochrane Library, SCOPUS and JSTOR. The search terms included "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title or text. The full search strategy is included in the supplementary appendix section one, and we apply the same search criteria for all the literature databases. The initial search included material published up to 28th May 2020. Studies were limited to those in which the subjects were human. Relevant reports and literature cited in the papers identified, as well as material from other sources such as the COVID Response Monitor developed by the European

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Observatory on Health Systems and Policies were also considered.[10] The initial search was subsequently updated to include material published between June 2020 and January 2021 (supplementary section two).

Study selection In the MEDLINE search, clinical studies, clinical trials, evaluation studies, government documents, journal articles, multicentre studies, observational studies, practice guidelines, pragmatic clinical trials and technical reports were included, as were reviews. Studies whose subject areas were in medicine or social sciences in SCOPUS, and research reports in JSTOR (excluding book chapters) were selected. We screened title, abstract and content to remove studies unrelated to elements of FTTIS or COVID-19.

Data collection process and data items For each publication we extracted (where appropriate) information on study design, method, results, main findings, and limitations using the PICOS tool.[11] **Summary measures and synthesis of results** Following discussion with researchers involved in the COVID-19 response in the UK (https://www.independentsage.org/), a list of key questions to answer was developed and answered using the literature collected. Sheng-Chia Chung, Sushila Marlow and Nicholas Tobias screened studies and extracted the data and disagreements were resolved through discussions.

Risk of bias in individual studies As studies eligible for policy review were mostly qualitative, we managed by using the Critical Appraisal Skills Programme (CASP) tool to assess the methodological rigour of included studies[12] (CASP Qualitative Studies Checklist).[13]

Patient and Public Involvement The study questions were inspired by weekly public discussions held by Independent SAGE, a scientific group for COVID-19 response in the UK (https://www.independentsage.org/).

RESULTS

In the initial search, we identified 50 studies from PUBMED (Medline), 43 studies from SCOPUS, 56 from JSTOR, 1 from the Cochrane library and 4 from additional sources. The flow diagram is set out in Figure 1. From the 164 initial publications, we removed 30 duplicate studies, 84 items that were not directly focused on COVID-19 or strategy of testing, contact tracing or isolation, or not in the format of a research report (such as editorials or podcasts). We subsequently identified an additional 58 studies in the second supplementary systematic review discussing FTTIS strategies during the early phase of the COVID-19 pandemic published between June 2020 and January 2021. We thus included a total of 118 studies in the review (supplementary table S1). Eligible studies had clearly stated aims and applied appropriate methods, and although none addressed all proposed study questions, they provided data relevant to an aspect in our review (supplementary table S2). A full list of studies included in the review is listed in supplementary section three.

Overview of FTTIS

We initially review of some key findings on the COVID-19 pandemic. A review by Nussbaumer-Streit B et al. concluded that a combination of quarantine and other non-pharmaceutical interventions could significantly reduce COVID-19 incidence and mortality.[14] Policy decisions should be informed by a detailed understanding of transmission dynamics. With growing evidence that so-called super-spreading events played an important role in many outbreaks,[15] finding, testing and isolating those present at such events was recognised as a priority.[16] This relates to work by Manchein et al who proposed that the growth of COVID-19 cases follows a power-law pattern rather than an exponential one, suggesting that transmission is characterised by features of scale free networks, fractal kinetics, and small world features. The implication is that while social distancing has been important to slow the spread of COVID-19 so far, a robust testing and isolation would be necessary to facilitate economies to reopen.[17] These conclusions were consistent with work from Italy stressing the importance of maintaining very strict quarantine of those affected. [18] Policies must also be underpinned by high quality population level data, yet some countries have struggled to create well-functioning integrated systems. Inadequate testing will often underestimate the true number and growth rate of COVID-19 cases.[19]

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[20] This is a challenge because testing is at the heart of a comprehensive FTTIS system and the only way to make a diagnosis where symptoms are atypical or unclear or in cases that are asymptomatic or presymptomatic.[21] For incidence, analysing surveillance data found that the high case-fatality during the first 40-day of COVID epidemic in Italy was primarily due to insufficient testing to characterise the outbreak accurately, thus masking transmission in the community and healthcare facilities.[22] In countries where FTTI interventions have been implemented subsequently, there has sometimes been a delay in their effect on disease control.[23] There are now a number of reports of how FTTIS systems have been designed and operated.[24]

1. Infrastructure and procedure of a sampling-to-results process that is 24 hours or less.

Countries have differed in whether they have built on existing infrastructure or created new structures to deal with the pandemic. We have identified two complementary elements in the reviewed studies, one involves increasing sampling capacity and the other scaling up laboratory capacity. The UK addressed the first of these by implementing drive-through SARS-Cov2 testing[25] with patients being informed of their results within 24 to 36 hours. It created a separate network of new stand-alone laboratories. Lagier et al reported on SARS-Cov2 RT-PCR testing of repatriated French citizens during their initial response,[26] with time from sample to result reduced to 3 hours by prioritising extraction of RNA from samples. Replacing the RNA extraction step by performing RT-PCR directly on heat-inactivated or lysed samples could accelerate the process further.[27] Point of care testing could also reduce the time to results to less than 2 hours.[28]

Binniker has set out the steps that should be taken when scaling up activity in existing laboratories,[29] which has the advantage of using networks already in place, including transport and communications. Similarly, Gupta et al have described how laboratories in universities and clinics in India were converted into testing centres.[30] In Ethiopia, Abera et al reported utilising Malaria and Neglected Tropical Diseases laboratories for COVID-19 testing.[31] Tolia et al described the implementation of a testing programme within a hospital emergency department.[32]

Strategies to improve the testing response

The case definition of potential COVID-19 changed frequently in the initial stage of the epidemic, as knowledge of the disease evolved. In the early stage of the epidemic, test kits and laboratory capacity were in short supply in many countries.[33] An example of an early investigation is when the Dutch National Outbreak Management Team investigated a potential SARS-Cov-2 local transmission, with rapid testing of health care workers in an area with a suspected COVID-19 outbreak. Testing followed a national protocol and was carried out either locally or in central laboratories. Nine hospitals were asked to sample health care workers on Saturday, 7th March and results were due on Monday, 9th March. When the SARS-Cov-2 positive rate came back as 4.1% among 1,097 tested, the local outbreak was confirmed.

Other laboratory-based strategies to improve testing efficiency include Yan et al. suggesting placing samples in reagents containing guanidine salts, for example, TRIZOL, TRIZOL LS, or AVL buffer, to inactivate the virus and protect RNA.[34] Pooled testing approaches have been advocated to increase testing capacity[35 36] and facilitate the detection of asymptomatic cases.[37] Only if the pooled batch tested positive would samples contributing to the batch be tested individually.[38] A three-stage pooling system was reported by Eberhardt et al who showed that it was possible to test three and seven times as many individuals with the same number of test kits when prevalence rates were 5% and 1%, respectively.[39]

Several studies examined strategies to test high-risk individuals,[40 41] concluding that it had potential if well-targeted,[42] whereas the utility of asymptomatic testing in low-prevalence setting may be low.[43] Targeted testing, measured by tests per confirmed case, was reported to be correlated with a greater reduction in COVID fatality than population testing number (tests per million people) in OECD countries and Taiwan.[44] One measure of the performance of a contact tracing and testing system is a shorter interval between the diagnosis of the index and secondary cases.[45 46]

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Minimising false test results.

Inaccurate results can impede both the treatment of the patient and public health interventions to contain COVID-19 infection.[47] Testing accuracy may be influenced by sampling and testing kit performance. The gold standard for COVID-19 diagnosis remains RT-PCR–based assays with respiratory specimens, although these can give positive results due to the persistence of fragments of genetic material after the individual has ceased to be infectious.[48] Among different respiratory sampling methods, RT-PCR assays based on nasopharyngeal swabs tend to have a lower cycle threshold than oropharyngeal swabs.[34] Infected individuals who are initially RT-PCR negative are likely to become positive with repeated tests as the infection progresses,[49 50] reflecting differences in the amount of virus at the site samples and viral shedding at the time of sampling.[49-51] Sampling error can occur due to low levels of virus at the beginning or end period of shedding or inadequate sampling techniques. Thus, recommendations are to take multiple samples,[49 51 52] while some have proposed supporting RT-PCR results with Chest Computerized Tomography imaging .[48 53]

The performance of tests has improved over time. Thus low sensitivity (30%) was reported in the RT-PCR testing kits in the early stage of the pandemic.[54] In a review by Younes et al, RT-PCR testing kits developed in the US, France and Germany had a high sensitivity, at 95% but specificity was not reported.[48] Yan et al have suggested that human RNase P gene could be amplified as an internal control to reduce false-negative results and template volume could be increased to improve sensitivity.[34] One study from March 2020 in India found no false negatives but only 5 in 7 positive tests were confirmed.[30] As a consequence, all apparent positive tests were retested by the National Institute of Virology (NIV). Point of care rapid testing has obvious benefits in terms of speed of results[55] but Döhla et al reported that tests available in the initial period of the pandemic had low sensitivity (36.4%) compared to RT-PCR, although this was in a small group of 39 COVID-19 patients.[51]

Besides RT-PCR testing, serologic antibody assays detect immunoglobin G and immunoglobin M to SARS-CoV-2 after infection.[56] This has been used to determine if an individual or the proportion of a population have previously been infected.[57] Given concerns about the duration of antibodies and limitations in sampling and testing, there is a need for caution about potential false negative results, and repeated testing may be required if symptoms persist. Testing at the end of the quarantine may help identify and care for delayed or asymptomatic COVID-19 cases while preventing further transmission.[58]

In addition to the aforementioned challenge relating to false negative-results, concern has been raised about the specificity of rapid antigen assays, with the risk of a high proportion of false positives, particularly as overall population prevalence reduces. For this reason, some countries are instigating PCR confirmation of positive rapid test results.[59]

Linked data to facilitate active case finding

There are several examples of how linked health service data can be used to identify cases.[24 60] For example, the digital COVID-19 border control system in Taiwan collects the health information of inbound passengers. The data are linked to the "epidemic prevention tracking system" used by health authorities to ensure isolation and quarantine of cases and are linked to the National Health Insurance MediCloud System, a cloud-based platform for medical professionals to gain updated clinical data of the patient seeking care.[61] The data linkage provides medical staff with real-time information on the patient's COVID-19 travel and contact history, home isolation and quarantine status, to provide necessary care with appropriate infectious control measures.[62] Screening high-risk populations (such as health care workers) has been used for active case finding.[63]

2. Adherence to isolation and local support needs

Inbound traveller quarantine

Border controls and restricted entry were imposed in East Asian countries early in the pandemic,[54] linked to comprehensive inbound traveller quarantine. For example, Singapore imposed a 14-day "Stay Home Notice" (SHN) on visitors and returning residents from COVID-19 endemic areas since January 2020 and for all inbound traveller from 9th April 2020. Upon arrival, travellers underwent a 14-day mandatory stay in government-

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designated hotels. During this period, they were not allowed to leave their individual rooms, where specific infectious disease prevention procedures were implemented.[64]

Timely identification of contacts

Ferretti et al. reported an estimated reproduction number of the initial SARS-Cov2 variants of 2.0, of which 0.9 was associated with infection occurring during the pre−symptomatic stage, suggesting about half of the cases were infected by COVID-19 patients before symptom onset.[65] An individual infected by SARS-Cov2 can actively shed the virus from their respiratory system 6 days before and 3 weeks after displaying symptoms, depending on disease severity.[34 48] While the virus detected in respiratory samples peaked within the first week of the illness, virus shedding continues in stool samples and peaks in week 2 or 3.[34] It thus is essential to stop the transmission by rapidly finding potential cases and their contacts and quarantining.[66] One modelling study reported how minimising testing delay effectively reduced onward transmission.[67] Countries may apply an additional week of self-health management beyond the 2-week quarantine period.[24]

Modelling studies have indicated that strategies to reduce random community contacts are preferable to those based solely on reducing personal contacts,[40 68] the duration of quarantine can vary depending on the reduction of contacts reached by individuals,[68] and testing at the beginning and the end of quarantine.[69] The definition of contact may vary; Korea applied a focused testing strategy where, when a COVID-19 outbreak with an unknown source of infection occurred, there was the testing of individuals who were working, living, or visiting in the outbreak area.[70] Individuals who tested positive were hospitalised, whereas individuals who tested negative were self-quarantined for 14 days. The strategy effectively reduced the proportion of COVID-19 cases with an unknown source of infection and the risk of transmission in the community.[70] The direction of contact tracing may be forward or both forward and backward; the latter identifies unascertained or asymptomatic carriers and enables the isolation of those involved in otherwise undiscovered transmission.[28]

Digital tools to facilitate FTTIS

Conventional epidemiological contact tracing, which relies on personal interviews is labour-intensive and time-consuming and may be challenging during a large-scale epidemic.[24 71] Some countries have applied digital tools in COVID-19 responses. Measures that use automated monitoring of geolocation data generated automatically by smartphones have advantages over voluntary tools (such as Apps) but there may be differing concerns about privacy among countries.[72-74]

Types of telecommunication provider-based measures may include:

1. Mapping anonymized cell phone movement in a particular area (for example, Germany, Austria, Italy),
2. Base station triangulation to approximate cell phone location (Taiwan and Hungary[75]) or
3. Access to GPS data generated by the phone (Israel)

Voluntary provision of data includes:

1. An App that captures proximity to other enabled mobile devices within a specified range via Bluetooth (Singapore, Austria),
2. An App recording daily symptoms (South Korea, Taiwan, Poland, UK) or
3. A QR code for entry or exit to key locations (China).

Aslam et al reviewed the digital tools used in East Asian countries for COVID-19 response by their level of intrusiveness.[54] Low intrusive digital tools provide services to individuals and act as the interface between authorities and service users.[60] Such tools are generally widely accepted. Other tools are used to enforce quarantine and may involve individual movement restrictions and privacy concerns.[76] These are subject to debate and have lower acceptability, yet large scale uptake (60% to 75% of the population) may be necessary for them to be effective.[73] For example, in Singapore, the initial voluntary use of the contact tracing App was

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only 12%.[73] The use of such Apps requires a clear legal basis for the information which can be used exclusively for infectious disease control and comes with strong legal limits on data access.[54 74]

Support for individuals complying with self-isolation

Webster et al. identified factors associated with adherence, which included the knowledge people had about the disease and quarantine procedures, social norms, perceived benefits of quarantine and perceived risk of the disease, as well as practical issues such as access to essential supplies or the financial consequences of being out of work.[77] Self-isolation or quarantine may not be affordable for low-wage and informal workers.[78] To address financial needs, the Singaporean government provides self-employed persons and businesses with employees undergoing self-isolation or quarantine (SHN) a $100 per day support. Deliveries of food and other supplies can be arranged for individuals in SHN, through a designated hotline. By law, landlords and dormitory operators cannot evict tenants in SHN. For individuals in residences that may not be suitable for SHN, the government offers hotels as an alternative.[64]

Psychological stress,[77 79 80] cognitive dissonance[74] or lack of physical activities[81] during self-isolation may adversely affect mental or physical health. While anxiety was reported to be positively associated with an interest in security-related actions, perceived restriction to personal freedom related to a negative association and loneliness to nonaction.[82] In a rapid review, Brooks et al reported individuals in quarantine experiencing confusion, anger, and post-traumatic stress symptoms, for reasons such as fear of infection, frustration, boredom, and inadequate supplies.[77] Austria and Germany reported an increase in domestic violence associated with COVID restrictions.[83] Strategies to mitigate these adverse effects include timely and sufficient information to reduce uncertainty, minimising the quarantine period to no longer than is necessary, and providing adequate supplies for individuals serving self-isolation.

Razai et al. suggested using validated psychological screening tools, such as the UCLA Loneliness Scale and Patient Health Questionnaire-4 for anxiety and depression to identify individuals who need support. This can help identify those who may benefit from counselling, coaching via telephone or online video consultations.[80] Non-medical social prescribing, such as visual choirs, online theatre or performances, exercise or art classes can reduce boredom. Communication using social media may ease loneliness. Banskota et al reviewed smartphone Apps to assist older adults in coping with self-isolation, designed for social networking, food and drinks delivery, medical consultation, health and fitness.[84] Meinert et al described an agile process to develop an App for older people and their families to improve well-being while observing social distancing rules.[85]

3. How would real-time data management, linkage of datasets, and dashboards be developed, and who would "own this"?

In Austria, the nationwide online Covid-19 dashboard has facilitated effective early response to COVID pandemic.[83] In South Korea, the government links digital databases to facilitate contact tracing, including electronic health records, phone-based GPS, card transaction records, and closed-circuit television.[72] Yasaka et al have proposed the use of an App with three guides, requiring users to register with checkpoints (e.g. public spaces or shops with a QR code that can be scanned to the app), check risk level, and report COVID-19 status.[86] Likewise, in China, a QR code-based App is used to limit the movement of suspected COVID-19 patients and displays a green, amber, or red code that is required to be in or enter locations, but the digital surveillance architecture and data protection mechanisms remain unclear.[54 73 74] Consequently, public concerns about data protection and privacy have been raised in respect to both the Chinese and South Korean Apps.

The efficacy of mobile positioning data (although not possible for 2G phones) was studied in Nigeria by Ekong et al. based on systems used in South Korea, Singapore and China.[87] A legal framework for data protection was suggested when implementing these systems and a third-party agreement was put in place for data use.

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Mobile geopositioning data

Mobile geopositioning has been used to study the mobility, connectivity, and health risks in travellers.[88] These have an accuracy of 150 meters so they reduce the risk of undermining individual privacy.[24] Taiwan has applied a geopositioning method to facilitate adherence to home isolation or quarantine, rapidly identifying contacts in a suspected large-scale outbreak. An example is the identification of 627,386 contacts of the 3,000 Diamond Princess passengers touring in Taiwan during a COVID-19 outbreak.[24] The process took a day and text messages advising self-health management were sent to all contacts. Contact data were linked to electronic health records for follow-up and testing of symptomatic contacts. Geopositioning data has lessened the pressure on health authorities who would otherwise have relied upon resource-intensive manual contact tracing, facilitating timely large-scale outbreak containment. The resources saved by leveraging technology can then be used for taking care of vulnerable populations and those without access to a mobile phone.

4. How would a "rapid response" occur, and what would precipitate such a response?

On 22nd January, an individual from Wuhan who had travelled to Hong Kong reported respiratory symptoms, and a second suspected case was reported the next day. Both received medical care, being placed in isolation, and later testing positive for COVID-19. Contact tracing started immediately and the travel histories of patients were retrieved and published online. All their close contacts, including passengers seated close by and taxi drivers, were subject to quarantine at a holiday village converted to a quarantine centre to host contacts of confirmed cases. A hotline was set up to answer public enquiries, especially for passengers on the same train/flight.[85]

In Singapore, proactive contact tracing and cluster identification are key features of the country's COVID-19 response. Health professionals are trained to identify potential outbreak clusters, asking a series of questions to COVID-19 patients. The Ministry of Health works with hotels to develop quarantine sites and liaises with those possessing CCTV footage to track cases.[85]

Infection control

Inadequate infection control during isolation or quarantine may increase the transmission of SARS-Cov-2. Close quarter isolation was found to be ineffective [89] in the case of the Diamond Princess. The virus spread to 634 passengers despite contact tracing[90] and safety measures,[91] leading the authors to conclude that aerosol transmission played an important role in confined settings. However, this was questioned by Wang et al in a hospital study of transmission via air, sewage, surfaces and personal protective equipment in which swabs tested positive only from sewage.[92]

Nosocomial infection and cross infection of health workers were reported at the Sheffield Teaching Hospitals NHS Foundation Trust; of 1,533 symptomatic healthcare workers, 18% were positive for SARS-CoV-2.[93] It was estimated that a third of staff had completed a shift while symptomatic, emphasising the need for regular and efficient testing for healthcare workers who have a high risk of infection, for the protection of vulnerable patients and civilians. A study in Taiwan on the risks of contact infection before and after symptom onset showed that the spread of COVID-19 was highest within the first 4 days of infection. In comparison with transmission to medical staff, most transmissions took place in the family or social settings, partially due to the absence of PPE and partially due to contact in the early stages of infection (pre-symptomatic).[94]

To control nosocomial infection, Taiwan implemented nationwide enhanced Traffic Control Bundling (eTCB) in hospitals whereby infection was controlled with a combination of triage prior to hospitalisation, separation between risk zones, strict PPE use and hand disinfection checkpoints.[60 95] Risk zones were divided into the contamination, intermediate and finally, clean. This drew on earlier research by Yen et al in which SARS infection among HCWs was transmitted to 2HCWs (0.03 cases per bed) in the eTCB hospital compared with 50

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probable cases (0.13 cases per bed) in the control group.[96] This strategy was implemented across Taiwan on 21st May, 2003 and within two weeks the epidemic was under control.

5. How would an App be assimilated in light of the above?

The Singaporean government has developed the App "TraceTogether" to support manual contact tracing,[97] recording other users who have been in proximity to a smartphone user via Bluetooth. After a user is found to be positive, individuals at risk are contacted directly.[54 74] The EU PEPP-PT coalition proposed a privacy-friendly contact tracing apps with the use of matching Bluetooth signals, based on the Singaporean "TraceTogether" app.[73] The Korean Ministry of the Interior and Safety has developed a mobile phone application named "self-quarantine safety protection" App that monitors the location of the quarantined user, informs health authorities, allows the user to report on their symptoms, and health officials can evaluate if a test is needed.[72] TRACE, the national contact tracing program in Taiwan, developed in 2017 has an integrated system to monitor health, trace contacts and link to other databases for a range of infectious diseases. The location of individuals could be tracked and communication between the user, Taiwan CDC, police and authorities as well as local health departments could be centralised through this platform for reporting symptoms and contact tracing.[98]

6. What are the barriers to & enablers of being tested, reporting contacts & isolating as a result of being contacted?

Logistics A lack of public health funding may weaken outbreak preparedness.[99] Regionalisation of the healthcare system might result in fragmentation of the epidemic response and timely data availability.[22] Scarcity of healthcare resources creates major barriers to COVID-19 control.[100]

Testing With the United States in mind, Parmet et al. encouraged the promotion of "free testing" in order to reach poorer communities and eradicate the infection.[78] Although the UK implemented free testing for adults, there have been challenges in communicating information on how to obtain a test and in what circumstances, particularly among certain ethnic minority groups. Understanding the factors associated with testing uptake in the population may inform better testing strategies.[101] Mark et al. reviewed the feasibility of a mobile community testing team in Scotland and reported a lack of guidance on infection control for testing in the community, thereby risking cross-transmission. Other barriers included the shortage of staff,[102] shortage of testing kits and strict testing eligibility which excludes asymptomatic patients from receiving testing in time.[83]

Contact tracing Validity and reliability of information recorded by an App may not be accurate or precise due to technological limitations.[84] Klonowska et al summarised the high coverage needed for an App to be effective. This may be difficult to reach on a voluntary basis, for issues such as mobile storage data, operating systems, and battery power to support constant Bluetooth activation, the fact that young children and senior citizens may not carry or own a personal smart device, and the issue of individual willingness.[73] In the UK, only 47% of individuals who are 75 years or older use the internet.[103] Senior citizens in the Netherlands were also less likely to adopt a contact tracing App than their younger peers.[104] It is essential that outbreak control teams cater to their needs and those of community volunteers, such as the NHS Volunteer Responders in the UK.[80]

Isolation Logistic challenges arise in providing food, sanitation, transport[105] and care for individuals living in a restricted zone, especially if this covers a large area.[78]

Public awareness and Communication

The number of tests carried out during an emergency will depend on public perception of the reliability of testing services and the effectiveness of communication of actions that can be taken. It is also necessary to ensure that kits and the location of testing stations are accessible to disadvantaged groups who may not have

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access to a smartphone or basic supplies. Where ethnic minorities are at greater risk, special efforts must be made to encourage them to access tests. Regular press conferences by authorities, briefing on the progress of the pandemic, changes in policies, and correcting misinformation[60] can increase public awareness of the need for FTTIS and tackle rumours that arise where there are knowledge gaps and uncertainties.[95] However, underreporting of COVID deaths by the government may occur in countries with incomplete vital registration or under political influence.[20 106]

The internet and social media have an important influence on behaviour in isolation,[107] which has grown with increased internet use since the COVID-19 lockdown. Farooq et al. tracked the effects of social media, news websites and emails as well as the living situation on the individual-level, and the intention of self-isolation during the pandemic in a panel of 225 participants,[108] and found that while frequent social media use contributed to information overload and cyberchondria, it increased propensity for self-isolation. McNeill et al similarly studied the effects of tweets and found that social media played a role in the motivation to conform to health measures.[109] Based on populational surveys, Kowalski et al reported that conspiracy beliefs of COVID-19 were negatively associated with adherence to self-isolation guidelines in Poland.[110] To motivate members of the public to self-isolate in a healthy way, a combination of lowering perceived response costs and clear information about the severity of risks should be implemented.[108 111] It is also suggested that messages of the severity of disease be disseminated via reliable official networks such as government outreach, news and journalism, while social media focuses on hopeful messages and recommended health measures.[112]

Legal issues

In Singapore, the Infectious Diseases Act provides a legal basis to prosecute those who are non-compliant, with resulting fines and/or imprisonment.[64] Similarly, the Communicable Disease Control Act in Taiwan mandates the government to implement effective measures in COVID-19 controls including FTTIS.[60]

In the US, the Public Health Service Act empowers the Surgeon General (delegated to the Centers for Disease Control and Prevention (CDC)) to enforce quarantine and isolation and provide medical care for detained individuals. Court rulings establish that quarantine cannot be imposed in a way that is racially discriminatory, and governments are required to have a strong legal basis for imposed restrictions.[78]

Public cooperation

The effective control of the highly contagious COVID-19 epidemic relies on collaboration by individuals to voluntarily provide their information and comply with preventive measures. In low or middle-income countries, low public trust due to government corruption impedes public cooperation to observe FTTI regulations.[113] The potential undesirable outcome of telling the truth, such as adverse immigration consequences for noncitizens, deportation of undocumented migrants, or refusal of entry for passengers with a fever, can be a barrier for these individuals, who may be at high risk of infection, to seek care or provide accurate information.[78] Results from a multinational survey in France, Germany, Italy, the UK and United States showed high public acceptability of a contact-tracing app.[114] Perceived benefits, self-efficacy[115], and trust[116] were positively associated with contact tracing apps uptake. Special protection of personal information is needed, with the goal of facilitating trust, prioritising care, and encouraging individuals to cooperate, to prevent further transmission of the disease.

Economic support

To ease the economic pressure and uncertainty during large-scale quarantine, governments have established different schemes to support individuals and families. Income compensation during quarantine has shown to double the compliance with self-isolation.[117] East Asian countries have provided support schemes for employers, employees and households.[54] In the United States, the federal government passed the Families

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First Coronavirus Response Act, including paid sick leave and unemployment insurance.[96] Countries implemented funding, financial stimulus, loans, aid packages to support the economy[83] or individual livelihood.[118] In the UK, national job retention schemes and financial support schemes have been rolled out to support individuals and businesses during the COVID-19 epidemic.[96]

Discussion

We identified in our systematic review the core elements for an effective Find, Test, Trace, Isolate, Support (FTTIS) system necessary to interrupt the spread of a novel infectious disease, as in the COVID-19 pandemic. The optimal strategies for a successful FTTIS during COVID-19 rapid response include:

Finding Border controls, restricted entry, and inbound traveller quarantine applied early in the pandemic have both reduced the spread of COVID-19 and facilitated case-finding. Testing high-risk populations (such as health care workers or residents of long-term care facilities) is important for active case finding, especially during the early stages.

Testing During the initial response to a pandemic, re-purposing of an existing laboratory network for testing should be implemented where possible while establishing new testing sites to meet increasing needs as the pandemic progresses. Pooled testing can be used to improve testing efficiency. Repeated testing may be necessary to minimise false-negative results.

Trace and Isolate Digital tools such as apps, GPS, and mobile geopositioning data have been deployed effectively by some countries for contact tracing and to monitor compliance with self-isolation. As an individual infected by SARS-Cov2 can actively shed the virus 3 weeks or longer after symptoms onset, the duration of quarantine may vary. A two-week quarantine period with extension when required has been adopted by various countries.

Support can be in the form of mental health or physical health promotion or monetary aid, such as timely and adequate information to reduce uncertainty and anxiety, providing adequate supplies or allowance for individuals. These support measures also facilitate public trust that is fundamental in the joint COVID-19 response in the population, as it relies on individuals to voluntarily provide their information and adhere to disease control regulations.

Integrated system Although these measures, often combined, are implemented in different countries, the level of preparedness, timely implementation and scale-up of effective measures made the difference between [119] or within[120] countries. For incidence, insufficient testing and quarantine led to the undetected wide transmission of SARS-Cov-2, resulting in high case-fatality during the first month of COVID epidemic in Italy,[22] and similarly in the Middle East[20] or during the first wave in the UK.[121]

A successful FTTIS requires seamless integration of all the components and the ability to integrate vulnerable populations or ethnic minorities. Linked health service data can contribute to the cooperation and communications among responsible authorities (central and local government, borders, laboratories and the medical care system) through real-time data sharing and dashboards.

Public collaboration In addition to support for individuals, regular press conferences by the central outbreak control team to provide a briefing on the progress of the pandemic, changes in policies, and correcting misinformation can increase public awareness of the need for FTTIS. Open and balanced discussions on public concerns, such as personal data privacy and protection and limitations on individuals freedom for the public good, are essential.

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Agile process The uncertainty and rapid progression of the pandemic make it important to have the flexibility to adapt rapidly to emerging challenges. A rolling-wave approach to planning can help authorities learn, apply and refine tools for prevention and control, with an agile process allowing better responses to emergencies.

Strengths We found no published systematic review addressing simultaneously the finding, test, tracing, isolation and support in the early response of COVID-19. Results of our review thus may inform countries facing future pandemics.

Limitations We reviewed studies published in English thus enable the review team to cross-validate the results. Although information documented in other languages might be missed, our search included studies from all regions of the world. Our review focused on FTTIS in the early phase of the pandemic. However, going forward, it will be important to draw on lessons learned later in the pandemic.

Conclusion

This comprehensive systematic review identified effective strategies for a successful FTTIS system to interrupt the spread of a novel infectious disease. These include border controls, restricted entry, inbound traveller quarantine, and screening for case-finding; repeated testing to minimise false diagnoses and pooled testing in resource-limited circumstances; extended quarantine period and the use of digital tools for contact tracing and self-isolation. Support for mental or physical health and livelihood are needed for individuals undergoing self-isolation/quarantine. An integrated system with rolling-wave planning can best utilise effective FTTIS tools to respond to the fast-changing COVID-19 pandemic. Results of the review may inform countries considering implementing these measures.

Contributorship

SCC, SMa, NT performed the acquisition, analysis, and interpretation of data for the work and draft the manuscript. All authors (SCC, SMa, NT, AA, IA, S-LY, KK, MM, SM, DP) contributed to the conception and design of the work, further data interpretation, revising it critically for important intellectual content, provided final approval of the version to be published and agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. SCC had final responsibility for the decision to submit for publication.

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Declarations of interest

The authors declare no conflict of interests.

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Data sharing

Data are available in a public, open access repository. All data relevant to the study are included in the article or uploaded as supplementary information

Ethical Approval Statement

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Ethical Approval Statement is not applicable in the study.

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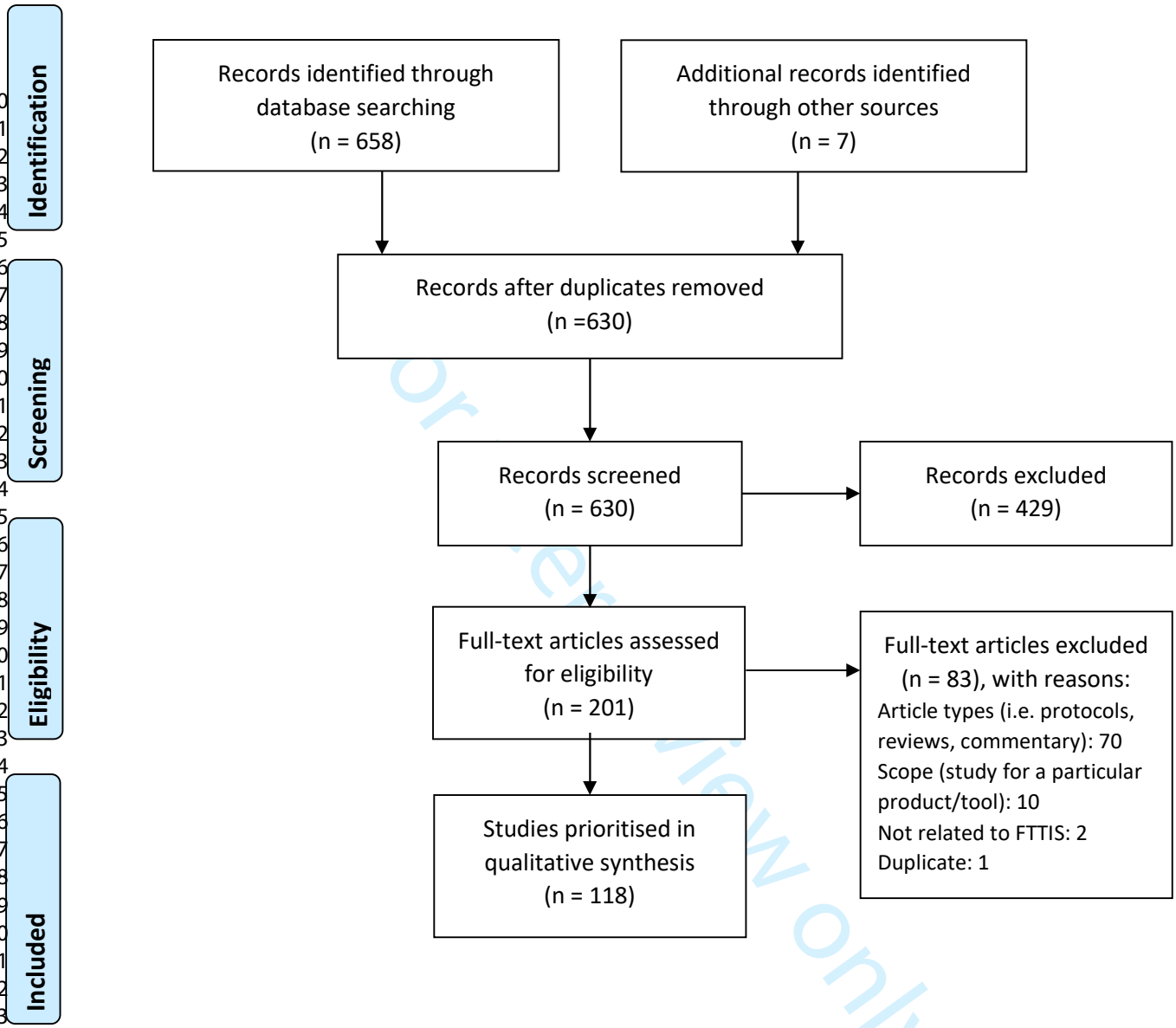
Figure Legend

Figure 1: Systematic review study flow diagram

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Figure 1: systematic review study flow diagram



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Supplementary appendix for the study

Lessons from countries implementing find, test, trace, isolation and support policies in the rapid response of the COVID-19 pandemic: a systematic review

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Supplementary section one: Search strategy

Excample database: MEDLINE (PubMed)

We applied search terms "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title or text. The studies were published between May 31th 2019 and May 31th 2020.

MEDLINE (PubMed) search: "contact"[Title] OR "contact tracing"[Title] OR "testing"[Title] OR "isolation"[Title] OR "quarantine"[Title] OR "self-isolation"[Title]) AND ("Covid-19"[Title/Abstract] OR "COVID-19"[Title/Abstract] OR "coronavirus"[Title/Abstract])

Filters applied: Clinical Study, Clinical Trial, Controlled Clinical Trial, Evaluation Study, Government Document, Journal Article, Meta-Analysis, Multicenter Study, Observational Study, Practice Guideline, Pragmatic Clinical Trial, Randomized Controlled Trial, Review, Systematic Reviews, Technical Report, in the last 1 year, Humans, English, MEDLINE.

We applied the same search strategy for Cochrane Library, SCOPUS and JSTOR.

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Supplementary section two: supplementary search strategy for the second review between June 2020 and January 2021

We applied search terms "contact tracing" or "testing" or "self-isolation" or "quarantine" in the title in combination with "COVID-19" or "COVID-19" or "coronavirus" in the title. The studies were published between June 1st 2020 and January 21st 2021.

Excample database: MEDLINE (PubMed)

(("contact tracing"[Title] OR "testing"[Title] OR "self-isolation"[Title]) AND ("Covid-19"[Title] OR "COVID-19"[Title])) Filters: Clinical Trial, Controlled Clinical Trial, Evaluation Study, Government Publication, Journal Article, Meta-Analysis, Observational Study, Practice Guideline, Pragmatic Clinical Trial, Randomized Controlled Trial, Review, Systematic Review, Technical Report, Humans, English, MEDLINE, from 2020/6/1 - 2021/1/21

Result in 392 studies

Exclude 73 studies have citations not yet completed MEDLINE indexing.

Result in 319 studies

Other exclusion criteria:

Items that were not directly focused on COVID-19 or strategy of testing, contact tracing or isolation in the general population, or not in the format of a research report, and studies with very small sample size <=5

After title screening: remove 180, 139 remains

After abstract screening: remove 49, 95 remains

We applied the same search strategy for Cochrane Library, SCOPUS and JSTOR.

The total studies after abstract screening: 148

The total studies after text screening: 58

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Table S1: Baseline table for reviewed studies.

Authors	Published year	Study type	Study aim(s)
Abdalhamid B et al.	2020	Laboratory report	To establish the optimal parameters for group testing of pooled specimens for the detection of SARS-CoV-2.
Aslam et al.	2020	Policy review	To analyse and provide policy recommendation for COVID-19 response
Banskota S et al.	2020	Narrative review	To narratively summarise 15 apps for older adults during quarantine
Binnicker MJ et al.	2020	Brief report	To provide suggestions on robust and sustainable system for testing
Brooks SK et al.	2020	Rapid review	Review of evidence on the psychological impact of quarantine
Canova V et al.	2020	Rapid assessment	To report on an urgent assessment of possible community transmission
Chen CM et al.	2020	Case study	To report on mass contact tracing and quarantine with big data
Döhla M et al.	2020	Technical report	To evaluate COVID-19 rapid testing methods
Eberhardt JN et al.	2020	Case study	To report on multi-stage group testing schemes
Ekong I et al.	2020	Rapid review	To exploratory review of COVID-19 contact tracing methods
Farooq A et al.	2020	Online survey	To evaluate the impact of online information on the individual-level intention to voluntarily self-isolate
Favaloro EJ et al.	2020	Commentary	To summarise a list of contemporary laboratory tests
Ferretti L et al.	2020	Simulation study	To model key parameters for COVID-19 transmission
Gupta N et al.	2020	Case study	To describe laboratory response to COVID-19 in India
Han E et al.	2020	Brief report	To summarise of COVID-19 response in Taiwan
Hellewell J et al.	2020	Simulation study	To develop a stochastic COVID-19 transmission model
Hill K et al.	2020	Technical report	To describe the drive-through coronavirus testing
Kakimoto K et al.	2020	Outbreak investigation	To investigate COVID-19 transmission on a cruise ship during quarantine
Keeley AJ et al.	2020	Technical report	To communicate results of the roll-out of SARS-CoV-2 testing
Klimburg et al.	2020	Policy review	To provide recommends digital contact tracing and quarantine tools.
Klonowska et al.	2020	Policy report	To report on the digital COVID-19 response in the EU
Koh D et al.	2020	Brief report	To discuss COVID-19 case finding
Lagier JC et al.	2020	Case study	To report on COVID-19 testing for repatriated flights passangers

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Lei L et al.	2020	Survey study	To assess anxiety and depression associated with COVID-19 quarantine
Manchein C et al.	2020	Modelling study	To analyse the mode of COVID-19 transmission
Marcel S et al.	2020	Case study	To describe the COVID-19 testing strategy in Switzerland
Mark K et al.	2020	Case study	To describe COVID-19 testing in Lothian, Scotland
Meinert E et al.	2020	Case study	To describe an app to mitigate isolation due to social distancing
Nussbaumer-Streit B et al.	2020	Rapid review	To implement a rapid review on the effectiveness of quarantine during COVID-19
Omori R et al.	2020	Modelling study	To model COVID-19 transmission
Parmet WE et al.	2020	Brief report	To discuss the legal aspects in COVID-19 quarantine
Peçanha T et al.	2020	Brief report	To discuss the cardiovascular burden of physical inactivity during COVID-19 self-isolation
Razai MS et al.	2020	Brief report	To discuss the psychological burden associated with COVID-19 self-isolation
Schulman R et al. .	2020	Brief report	To discuss the false knowledge associated with COVID-19
Schwartz J et al.	2020	Case study	To explore whether enhanced cooperation with local government and community institutions can effectively supplement state-centric public health epidemic responses.
Sjödin H. et al.	2020	Simulation study	To model the COVID-19 progression under different quarantine scenarios
Taghrir MH et al.	2020	Policy review	To review the effect of mass quarantine on COVID-19 in China.
Tang B et al.	2020	Simulation study	To model COVID-19 transmission
Tolia VM et al.	2020	Evaluation study	To evaluate COVID-19 testing program
Wang J et al.	2020	Outbreak investigation	To investigate route of COVID-19 transmission during a hospital outbreak
Webster RK et al.	2020	Rapid review	To identify and summarise factors associated with quarantine adherence
Wilder-Smith A et al.	2020	Narrative Review	To review of existing COVID-19 self-isolation measures and factors that influencing their efficacy
Xu JW et al.	2020	Technical report	To examine if isolation work in buildings with the example of the DIamond Princess as a case study
Yamahata Y et al.	2020	Case study	To describe COVID-response in Cruise Ship Diamond Princess outbreak
Yan Y et al.	2020	Narrative Review	To review COVID-19 laboratory testing methods
Yasaka TM et al.	2020	Case study	To develop an effective and non-intrusive contact tracing app

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Yen MY, Lin YE, et al.	2020	Case study	To report strategies for controlling a hospital outbreak
Yen MY, Schwartz J, et al.	2020	Case study	To report strategies to control a hospital outbreak
Zhang MX et al.	2020	Simulation study	To model the optimal transportation of COVID-19 patients to minimise subsequent transmission
Zitek T.	2020	Technical report	To discuss on the appropriate use of COVID-19 testing
McCombs A et al.	2020	Simulation study	To establish a network model of COVID-19 transmission to inform policy choices
Wang S et al.	2020	Case study	To describe a mini-program within the app WeChat for contact tracing
Altmann S et al.	2020	Online survey	To investigate the user acceptability of a contact-tracing app
Kowalski J et al.	2020	Survey study	To examine whether conspiracy beliefs have an adverse effect on COVID-19 medical guidelines adherence
Walrave M et al.	2020	Survey study	To study factors influencing the intention of using a contact-tracing app
Vicentini C et al.	2020	Retrospective study	To examine whether testing could be associated with variations in COVID-19 fatality.
Park Y et al.	2020	Case study	To describe COVID-19 testing strategy in Seoul
Nijsingh N et al.	2020	Technical report	To discussion on the reversibility of a contact tracing app
Edjoc R et al.	2020	Comparative study	To compare the mathematical tool to Rt estimates in COVID-19 epidemic in Canada.
Bradshaw WJ et al.	2021	Simulation study	To examine whether a bidirectional contact tracing can effectively identify COVID-19 transmission.
Magno L et al.	2020	Case study	To report COVID-19 testing strategy in Brazil
Rahaman KR et al.	2020	Case study	To describe COVID-19 testing strategy in Banglades
Falaschi Z et al.	2020	Evaluation study	To assess the computed tomography diagnostic accuracy against RT-PCR
Brendish NJ et al.	2020	Evaluation study	To assess the point-of-care testing in reducing time to results
DB S et al.	2020	Brief report	To describe COVID-19 response in different countries
Grassly NC et al.	2020	Simulation study	To assess different testing and isolation strategies on SARS-Cov-2 transmission
Cheng HY et al.	2020	Clinical investigation	To investigate COVID-19 transmission among close contacts
Jian SW et al.	2020	Case study	To report the COVID-19 contact tracing in Taiwan

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Wei C et al.	2021	Evaluation study	To compare COVID-19 population testing number and coverage on mortality
Eilersen A et al.	2020	Simulation study	To assess different COVID-19 response strategies through mathematical modelling
Ruebush E et al.	2021	Case study	To report on the COVID-19 contact tracing in the US
Jonker M et al.	2020	Simulation study	To model the uptake of a COVID-19 contact tracing app
Chanda A. et al.	2020	Modelling study	To study the COVID-19 transmission dynamics in India
Salameh P	2020	Evaluation study	To report COVID-19 epidemiology in countries of the Eastern Mediterranean Region
Brooks ZC et al.	2020	Simulation study	To model the effect of false-positive and false-negative COVID-19 testing results.
Mettler SK et al.	2020	Modelling study	To examine the indicator of diagnostic serial interval and its association with clinical onset of COVID-19
Malheiro R et al.	2020	Case series	To report the characteristics of COVID-19 patients in Portugal
Keeling MJ et al.	2020	Survey and modelling study	To model the effect of contact tracing on COVID-19 transmission based on network data.
de Wolff T et al.	2020	Mathematical modelling	To establish mathematical model to assess sample pooling method to enhance COVID-19 testing
Frowde R et al.	2020	Brief report	Critique on the COVID-19 response in the UK
Reiss S et al.	2020	Survey study and pathway analysis	To investigate psychological perceptions and defensive strategies towards COVID-19 and self-isolation.
Kretzschmar ME et al.	2020	Mathematical modelling	To identify factors for a succsddrul COVID-19 contact tracing
Smalley DL et al.	2021	Technical Report	To investigate the impact of pool testing and diluting RT-PCR testing reagent on COVID-19 testing.
Bodas M et al. Isr J Health Policy Res.	2020	Survey research	To assess public attitudes toward COVID-19
Li S et al.	2020	Cross-sectional survey	To investigate individual factors such as internet use and the uptake of COVID-19 prevention methods
Gray N et al.	2020	Modelling study	To assess strategies to end the lockdown for COVID-19 control.
Smyrlaki I et al.	2020	Technical report	To investigate the extracting method on rapid COVID-19 testing.
Bilinski A et al.	2020	Mathematical modelling	To assess contact tracing strategies within the ease of COVID-19 control measures.

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Aleta A et al.	2020	Mathematical modelling	To model the impact of testing, contact tracing and household quarantine on second COVID-19 wave.
Kaspar K.	2020	Survey research	To examine central cognitive variables influencing individuals' motivations for up-taking social distancing and apps for COVID-19 control
Wells CR et al.	2021	Mathematical modelling	To study optimal COVID-19 quarantine and testing strategies.
Huang Z et al.	2020	Evaluation study	To compare the performance of the contact tracing app and a wearable tag-based real-time locating system
Yap KY et al.	2020	Technical report	To present a COVID-19 web-app
Ezeibe CC et al.	2020	Qualitative study	To investigate the impact of political distrust on COVID-19 pandemic in Nigeria.
Gopaul R et al.	2020	Evaluation study	To assess the accuracy of COVID-19 nasopharyngeal swab PCR testing
Ferrari D et al.	2020	Evaluation study	To examine whether combining RT-PCR with routine blood test can reduce false-negative COVID-19 testing.
Bodas M et al. Health Aff (Millwood).	2020	Survey research	To assess public attitudes toward COVID-19
Deckert A et al.	2020	Simulation study	To simulate and compare results of the pooled-sample strategies for COVID-19 mass testing
Li Y et al.	2020	Evaluation study	To evaluate the consistency of COVID-19 RT-PCR testing results in hospitalised patients.
Chiu WA et al.	2020	Modelling study	To assess the effect of relaxation of COVID-19 control measures.
Indenbaum V et al.	2020	Evaluation study	To compare COVID-19 antibody testing to RT-PCR testing results
Müller M et al.	2020	Simulation study	To examine whether frequent sampling of non-symptomatic but infectious individuals in addition to testing symptomatic individuals can improve COVID-19 management
Xie K et al.	2020	Mathematical modelling	To study the mechanism of the risk perception effect on social distancing
Jung J et al.	2020	Case study	To study the impact of mandatory COVID-19 testing before releasing individuals from the 14-day quarantine in Incheon, Korea.
Hong K et al.	2020	Technical report	To examine the indicator of the Serial Interval of COVID-19 for contact tracing
Gul S et al.	2020	Simulation study	To investigate probability of COVID-19 transmission via different social network.
Sawano T et al.	2020	Case series	To report the factors associated with regional variation of COVID-19 incidence in Japan

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Jacofsky D et al.	2020	Technical report	To present a COVID-19 antibody tests
Szocska M et al.	2020	Case study	To report on using data on call detail records to monitor physical morbidity
Alger J et al.	2020	Evaluation study	To study the validity of COVID-19 antibody rapid tests
Schrager JD et al.	2020	Cross-sectional usability study	To test whether adult users could understand and recall the information provided by a COVID-19 web-based tool.
Bai AD et al.	2020	Point-prevalence study	To estimate the prevalence of COVID-19 among inhospital patients
Flynn EF et al.	2020	Case study	To report the design and implementation of COVID-19 drive-through testing for children and pediatric health care workers
Brendish NJ et al.	2020	Non-randomised clinical study	To assess the accuracy and effectiveness of point of care COVID-19 testing
Dinnes J et al.	2021	Systematic review	To assess the diagnostic accuracy of COVID-19 point-of-care antigen and molecular-based tests
Abera A et al.	2020	Case study	To report on establishing a COVID-19 testing laboratory in Ethiopia
Wu DC et al.	2020	Survey research	To report on the self-reported COVID symptoms from a national-representative poll in Canada
Younes N et al.	2020	Narrative review	To discuss laboratory and surveillance technologies for COVID-19 testing

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Table S2

Authors	Published year	1. Was there a clear statement of the aims of the research?	2. Is a qualitative methodology appropriate?	3. Was the research design appropriate to address the aims of the research?	4. Was the recruitment strategy appropriate to the aims of the research?	5. Was the data collected in a way that addressed the research issue?	6. Has the relationship between researcher and participants been adequately considered?	7. Have ethical issues been taken into consideration?	8. Was the data analysis sufficiently rigorous?	9. Is there a clear statement of findings?	10. How valuable is the research?
Abdalhamid B et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	Yes	Yes	Yes	Good
Aslam et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Banskota S et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Binnicker MJ et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Brooks SK et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Canova V et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Chen CM et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Döhla M et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Eberhardt JN et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Ekong I et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Farooq A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Favaloro EJ et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Ferretti L et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Gupta N et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Han E et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Hellewell J et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Hill K et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Kakimoto K et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Keeley AJ et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Klimburg et al. .	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Klonowska et al. .	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Koh D et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Lagier JC et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Lei L et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Manchein C et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Marcel S et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Mark K et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Meinert E et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Nussbaumer-Streit B et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good

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Omori R et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Parmet WE et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Peçanha T et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Razai MS et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Schulman R et al. .	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Schwartz J et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Sjödin H. et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Taghrir MH et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Tang B et al.	2020	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	N/A	Moderate
Tolia VM et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Wang J et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Webster RK et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Wilder-Smith A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Xu JW et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Yamahata Y et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Yan Y et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Yasaka TM et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Yen MY, Lin YE, et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Yen MY, Schwartz J, et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Zhang MX et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Zitek T.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
McCombs A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Wang S et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Altmann S et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Kowalski J et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Walrave M et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Vicentini C et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Park Y et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Nijsingh N et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Edjoc R et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Bradshaw WJ et al.	2021	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Magno L et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Rahaman KR et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Falaschi Z et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Brendish NJ et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
DB S et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Grassly NC et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Cheng HY et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good

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Jian SW et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Wei C et al.	2021	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Eilersen A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Ruebush E et al.	2021	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Jonker M et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Chanda A. et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Salameh P	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Brooks ZC et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Mettler SK et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Malheiro R et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Keeling MJ et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
de Wolff T et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Frowde R et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Reiss S et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Kretzschmar ME et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Smalley DL et al.	2021	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Bodas M et al. Isr J Health Policy Res.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Li S et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Gray N et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Smyrlaki I et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Bilinski A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Aleta A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Kaspar K.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Wells CR et al.	2021	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Huang Z et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Yap KY et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Ezeibe CC et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Gopaul R et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Ferrari D et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Bodas M et al. Health Aff (Millwood).	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Deckert A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Li Y et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Chiu WA et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Indenbaum V et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Müller M et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Xie K et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Jung J et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Hong K et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Gul S et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Sawano T et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Jacofsky D et al.	2020	Yes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Yes	Moderate
Szocska M et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Alger J et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good

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Schrager JD et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Bai AD et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Flynn EF et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Brendish NJ et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Dinnes J et al.	2021	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Abera A et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Wu DC et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good
Younes N et al.	2020	Yes	Yes	Yes	Yes	Yes	N/A	N/A	Yes	Yes	Good

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Supplementary section three: List of reviewed articles:

For the first review between May 2019 and May 2020

Klonowska, Klaudia, and Pieter Bindt. The COVID-19 Pandemic: Two Waves of Technological Responses in the European Union. Hague Centre for Strategic Studies, 2020.

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Klimburg A, Faesen L, Verhagen P, Mirtl P. Pandemic Mitigation in the Digital Age: Digital Epidemiological Measures to Combat the Coronavirus Pandemic. Hague Centre for Strategic Studies. 2020.

Parmet WE, Sinha MS. COVID-19 - The Law and Limits of Quarantine. N Engl J Med. 2020 Apr 9;382(15):e28. doi: 10.1056/NEJMp2004211. Epub 2020 Mar 18.

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Abdalhamid B, Bilder CR, McCutchen EL, Hinrichs SH, Koepsell SA, Iwen PC. Assessment of Specimen Pooling to Conserve SARS CoV-2 Testing Resources. Am J Clin Pathol. 2020 May 5;153(6):715-718. doi: 10.1093/ajcp/aqaa064.

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PRISMA 2009 Checklist

, GiSection/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	3,4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4, Figure
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	4
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	4
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	4
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	4
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	5
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	5
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	5
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	5-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	5-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	5
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	12
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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